



# **KENYATTA UNIVERSITY**

**DEPARTMENT: HISTORY, ARCHAEOLOGY AND POLITICAL STUDIES**

**AHT 212- EARLY AFRICAN ACHAEOLGY**

**Dr. Okelo**

**BY KELVIN EKALIYO, THE NEXT.**

# **SIR OTONDE MICHAEL AWITI-DELEGATE**

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## INTRODUCTION



This module introduces students to the archaeological regional geography and paleoenvironments. It provides a detailed description on modes for hominid evolution; early Stone Age technology and variation; the beginnings of culture and ecological specialization; Middle Stone Age technology and variation; origins of modern people in Africa; early Later Stone Age technologies; models of technological and economic change in Early, Middle and Later Stone Ages; and the origin of African art. The students of this module need to have taken AHT 101: Principles of Archaeology since it gives an introduction to basic understanding of archaeology and important concepts in archaeology.

## OBJECTIVES



At the end of this module, the student is expected to be able to articulate the following:

- a. The process and ways of reconstructing past environments
- b. Competently articulate why the African continent is important centre for explaining the evolution of humanity
- c. Have an understanding of how evolution of other animals relates to origin of modern humans.
- d. Have an understanding of the process of human evolution and how evolution is a subject to environment.
- e. Be able to explain the origin of genus homo and its spread throughout the world.
- f. Be able to articulate the archaeology of stone tools i.e. the Oldowan, Acheulian, MSA, and LSA.
- g. Articulate the origin and spread of African art

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## FOSSILS in GEOLOGICAL CONTEXT

### LECTURE ONE

#### 1.1 INTRODUCTION



In this chapter we will examine fossils in geological context, geological timescales and dating methods. The chapter will allow you to understand the process of fossilization and factors that are necessary for future recovery.

Over the last 50 years new techniques have been developed that allow scientists to provide a more accurate context of our understanding of the evolutionary past. To understand the past we study **fossils**. As preserved remnants of once living things, fossils provide information about the past life. **Palaentology**, a field that take its name from the Greek words for —oldl (*paleos*) and —existencel (*ontos*), is devoted to gleaning all the information that can be extracted from the fossils. A fossil without its context is almost useless because there is no way of assessing how old it is, what kind of environment it lived in, what other animals it might have lived around it. In order to answer the above issues, **geology** is applied. It is the study of the earth, which allows us to understand the preservation, age, and environment in which fossils are found.

#### 1.2 LECTURE OBJECTIVES



In this lecture students are expected to understand the process of fossilization, preservation, geological and biological processes necessary in the preservation and recovery of fossils. The lecture also provides on dating methods.

#### 1.3 HOW TO BECOME A FOSSIL

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In reality very few living things become fossils. Not only does a minute fraction of living things become preserved as fossils, but only an exceedingly small proportion of the fossils that are preserved end being discovered, collected and studied. Not surprisingly, then that the fossil record is not entirely representative of the composition of past biological communities ( Behrensmeyer and Hill 1980). Instead, fossil record preserves some organisms more than others. **Taphonomy** which is the study of what happens to organism from the time they die to their discovery is used. This approach may include both the biological and geological processes.

Death may come to the organism in a number of ways including disease, injury and or predation among others. In some cases, the agent of death such as predators may leave marks on the skeleton such as bite marks. After death the carcass begins to decompose when living tissues are no longer maintained by the organism, and numerous microbes such as bacteria and insects accelerate the decomposition process. As the decomposition goes on, scavenging animals such as hyenas may consume the soft tissues and in some cases even chew on the bones. Eventually, only the most durable tissues such as dense bones remain. However, in some cases even the dense bones may disappear due to erosion and trampling.

To become a fossil, part of the organism must be preserved by burial, a natural process by which the carcass or part of it is covered with sediment. Because sediments such as sand, silt, mud, and gravel usually are carried by water, burial often occurs in the floodplains of rivers, along the shores of lakes and in swamps where erosion and sedimentation are occurring. In other circumstance, sediments such as dust and volcanic ash carried by the wind sweeps over the remains. Once buried, skeletal remains may be preserved in variety of ways. Usually they absorb minerals from the surrounding soil or ground water that eventually replace the organism's original inorganic tissues. The result is *petrification* the process of being turned stone.

Fossilized remains usually preserve only the most durable tissues of the body, such as teeth and bones, and just as the inorganic component is replaced by minerals in ground water, eventually the organic component of the skeleton, what is referred to as collagen degrades and is lost. On occasion however, soft parts such as skin or hair may be preserved. In very exceptional circumstances, the original tissues of an organism are preserved, as when animal bodies are frozen, such as the mammoths of Siberia, or when ancient peoples are mummified. Presence of *coprolites*, or fossilized feces also tell about the presence of past animals.

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## 1.4 THE IMPORTANCE OF CONTEXT

As mentioned earlier, a fossil without its context is useless. In the following section we review the important principles used in geology to understand the position of a fossil in its rock layers and of different fossil sites relative to each other.

### 1.4.1 STRATIGRAPHY

Stratigraphy is the study of the distribution of layers in a soil. It is important since it allows archaeologists to place and locate the position of a find in the rock layers. In 1830 Charles Lyell, synthesized a number of accepted geological principles including the principles of stratigraphy. The principles of geology rely in large part on the concept of *uniformitarianism*, originally developed by James Hutton in 1785 and further promoted in Lyell's book. Uniformitarianism suggests that processes operating today are also those that operated in the past and thus they can explain the fossil and geological record. The principles of stratigraphy include four that are critical to understanding of the context of a fossil: Original horizontality, superposition, cross-cutting relationships, and faunal succession.

*Principle of original horizontality*, formulated by Nicolas Steno, points out that layer of rock( strata) are laid down parallel to the earth's surface, at least originally. Deformations might occur later due to activities such as earthquakes and volcanic eruptions. The *principle of superposition* states that with all the other factors equal, older layers are laid down first then overlaid by younger layers. Thus it means that sediments at the bottom and the artifacts found in them are older than the ones overlaying them.

However in some cases, stratigraphy is not always as easily laid out as layers of cake. Thus, the invention / proposal by James Hutton in the late 1700s of the *principle of cross-cutting relationships* which states that geological feature must exist before another feature can cut across or through it and that the thing that is cut is older than the thing cutting through it. The *principle of faunal succession* addresses the changes of animal remains (fauna) through layers. It was proposed by William Smith in 1815 and he recognized that the deeper fauna is older, and also that there are predictable sequences of fauna through strata, that successive layers contain certain types of fossils that follow one another in predictable patterns through the strata. Certain of these animals that typify a layer are called *index fossils*.

Thus, using the principles of stratigraphy, it can be determined which strata are older and younger? Comparisons between sites can provide a sequence of rocks from older to younger for both areas. According to Stanford, Allen and Anton (2006), by comparing stratigraphy of sites from around the world, especially for marine sediments that are continuous, geologists have assembled a great geological column from the oldest to the youngest rocks on earth. The geological column with age estimates provided by relative dating is called the **geologic time scale** (Fig. 1.1).

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## 1.5 GEOLOGIC TIME SCALE

The Geologic Time Scale (GTS) The geologic time scale provides a system of chronologic measurement relating to stratigraphy to time that is used by geologists and other earth scientists to describe the timing and relationships between events that have occurred during the history of earth. It provides an ordered, internally consistent, internationally recognized sequence of time intervals, each distinct in its own history and record of life on Earth, including the assignment of absolute time in years to each geologic period. The geologic time scale has a relative scale, consisting of named intervals of geologic history arranged in historical sequence; and a numerical (or absolute) time scale, providing absolute ages for the boundaries of these intervals.

The GTS is divided into nested sets of time. These are from the most inclusive to the least inclusive these are eons, eras, periods, and epochs. The earth is approximately 4.5 billion years old, and the GTS covers this entire time although human and primate evolution occurs in the Cenozoic era, about 65 million years.

The scale is divided into two eons, the Precambrian and Phanerozoic. The Precambrian dates from 4.5 billion to 543 million years ago and is divided into three eras: the Hadean, Archaen, and Proterozoic. The Phanerozoic Eon dates from 543 million years ago to the present and is divided into three eras Paleozoic, Mesozoic, and Cenozoic from oldest respectively. The Cenozoic spans from 65 million years ago to present and has two periods: the Tertiary and Quaternary.

The Tertiary Period, from 65 to 1.8 million years ago spans five epochs: the Paleocene (65- 54.8 million years ago), Eocene (54.8- 33.7 million years ago), Oligocene (33.7- 23.8 million years ago), Miocene (23.8 – 5.3 million years ago), Pliocene (5.3- 1.8 million years ago). The Quaternary period from 1.8 million years ago to present, spans two epochs: the Pleistocene (1.8 years ago to 10,000 years ago) and Holocene (10,000 to present).

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| EON         | ERA         | PERIOD        | EPOCH       | Ma   |
|-------------|-------------|---------------|-------------|------|
| Phanerozoic | Cenozoic    | Quaternary    | Holocene    | 0.01 |
|             |             |               | Pleistocene | 0.8  |
|             |             | Tertiary      | Pliocene    | 1.8  |
|             |             |               |             | 3.6  |
|             |             |               | Miocene     | 5.3  |
|             |             |               |             | 11.2 |
|             |             |               | Oligocene   | 16.4 |
|             |             |               |             | 33.7 |
|             |             |               | Eocene      | 28.5 |
|             |             |               |             | 33.7 |
|             |             |               | Paleocene   | 41.3 |
|             |             |               |             | 49.0 |
|             | Mesozoic    | Cretaceous    | Late        | 54.8 |
|             |             |               |             | 61.0 |
|             |             | Jurassic      | Early       | 65.0 |
|             |             |               |             | 99.0 |
|             |             | Triassic      | Late        | 144  |
|             |             |               |             | 159  |
|             | Paleozoic   | Permian       | Middle      | 180  |
|             |             |               |             | 206  |
|             |             | Pennsylvanian | Early       | 227  |
|             |             |               |             | 242  |
|             |             | Mississippian | Late        | 248  |
|             |             |               |             | 256  |
|             |             | Devonian      | Early       | 290  |
|             |             |               |             | 323  |
|             |             | Silurian      | Late        | 354  |
|             |             |               |             | 370  |
|             |             | Ordovician    | Early       | 391  |
|             |             |               |             | 417  |
|             | Cambrian    | Cambrian      | Late        | 423  |
|             |             |               |             | 443  |
|             |             |               |             | 458  |
|             |             |               |             | 470  |
|             |             |               |             | 490  |
|             |             |               |             | 500  |
| Precambrian | Proterozoic | Proterozoic   | Middle      | 512  |
|             |             |               |             | 520  |
|             |             |               |             | 543  |
|             | Archean     | Archean       | Late        | 900  |
|             |             |               |             | 1600 |
|             |             |               |             | 2500 |

Figure 1.1: Earth's history divided into Eons, Era Period Epochs referred to as Geologic Time Scale.

## 1.6 RELATIVE DATING TECHNIQUES

Relative dating is used to determine the relative ages of geologic strata, artifacts, historical events, etc. This technique does not give specific ages to items. It only sequences the age of things or determines if something is older or younger than other things. They tell us how something is in relation to something else without applying an actual chronological age. Majority of the relative dating techniques rely on geological principles of stratigraphy. These include lithostratigraphy, biostratigraphy, tephrostratigraphy, chemical methods, climate chronology, dendrochronology, ice core sampling, stratigraphy, and seriation..

**Lithostratigraphy** this uses characteristics of the rock layers to correlate across regions. *Litho* means rock. For example if a volcanic eruption spewed ash across a region such as the central rift valley from Naivasha to Baringo, then we would expect to find the ash layer even if they exhibit different sequences of rock layers above and below the volcanic ash layer.

**Tephrostratigraphy** this is the identification of volcanic ash by its chemical fingerprint of major, minor and trace elements. These chemical similarities allow us to correlate volcanic ashes (tephra)

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with each other demonstrating relation even in sites that are separated. This method has been used to identify and describe obsidian in the Rift Valley.

**Biostratigraphy** this method uses biological organisms found in rocks to correlate ages between sites. Thus, presence of given organisms tells the relative age of the sites, especially those that are known to have existed at a given period or those that died at a given time period. Rodents are often good biostratigraphic indicators of age in many regions. In the region certain species of pigs are good markers in the Plio- Pleistocene, since their extinctions are known

**Techniques within sites** some chemical techniques are useful for identifying the relative age of different fossils from the same sites. These techniques include the analysis of the fluorine, uranium, and nitrogen content of the fossils. These are particularly useful when the association between different fossils or between the fossils and their sediments is in question.

## 1.6.1 CALIBRATED RELATIVE DATING TECHNIQUES

**Calibrated relative techniques** include regular processes that can be calibrated to a chronological scale if certain conditions are known. Some of these techniques include obsidian hydration, amino acid racemization and paleomagnetism.

### **Obsidian Hydration**

This dating method works when dating obsidian based tools. It can be used to give relative date when comparing two or more artifacts, but could also be used to give absolute dates where a calendar date has been introduced. The dating method is premised on the fact that a fresh surface is created on a piece of obsidian in the tool manufacturing, process. Obsidian contains about 0.2 percent water. When fractured, atmospheric water is attracted to the surface and begins to diffuse into the glass. This results in the formation of a water rich hydration crust that increases in depth with time. The hydration process continues until the fresh obsidian surface contains about 3.5 percent water. This is the saturation point.

The thickness of the hydration crust can be identified in petrographic thin sections cut normal to the surface and observed under a microscope. A distinct diffusion front can be recognized by an abrupt change in refractive index at the inner edge of the hydration crust. These fronts or rinds of hydration are more dense than the unhydrated inside, and the unhydrated zone has different optical properties. Thus, the degree of hydration observed on an obsidian artifact could tell archaeologists how long it had been since that surface was created by a tool maker.

### **Racemization**

Racemization dating is a process which uses the measurement of the decay rate of carbon protein amino acids to date once-living organic tissue. All living organisms have protein; which is made up of amino acids. All but one of these amino acids (glycine) has two different chiral forms (mirror images of each other). While an organism lives, their proteins are composed of only 'left-handed' (laevo, or L) amino acids, but once the organism dies the left-handed amino acids slowly turn into right-handed (dextro or D) amino acids. Once formed, the D amino acids themselves slowly turn

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back to L forms at the same rate.

The ratio between the D and L versions of each amino acid increases with time from 0 (0%D / 100% L) to 1 (50%D / 50%L). This process is called amino acid racemization, and derives from the term racemic mixture, the point at which there is an equal balance of D and L amino acids within an organism. Once an organism has achieved a racemic mixture, even though L amino acids still racemize to D and vice versa, the D/L ratio will not change because the amino acids going in each direction cancel each other out. Racemic mixtures of amino acids have been found in fossils, millions of years old. Racemization can be used to date objects between 5,000 and 1,000,000 years old.

**Geometric polarity** currents in the earth's outer core create magnetism, and they change, the Earth's polarity may flip. At times in the past, magnetic north has been the opposite or reverse of today, which is in the South Pole. Such reversals occur quickly, perhaps over thousands of years, and do not last for set periods of time. During the process of rock formation, magnetic minerals orient themselves towards magnetic north. Rocks formed today would have a polarity or oriented to today's magnetic field.

Over the years Paleomagnetists have assembled a **geomagnetic polarity time scale (GPTS)** that records the orientation of sediments from different intervals. The time scale is based on a sequence of changes in the magnetism of ancient layers, or **paleomagnetism**, that has been pieced together, mainly from sediments that have spread outward from mid-oceanic ridges at the bottom of the sea (Cande and Kent 1995).

## 1.7 SUMMARY



The process of fossilization requires some very necessary occurrences and environment in order to occur. The process is subject to various biological and geological process that determine if whether when an organism dies it is recovered. The context that the organism dies and is preserved in is also very important in order for the organism to be recoverable in the future. Dating gives the fossil a time history in order to have a better understanding of the past.

## 1.8 SELF- TEST QUESTIONS



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1. Why is the fossil record so fragmentary?
2. What is taphonomy?
3. What is the difference between relative and chronological dating techniques?
4. How do scientists think climate changed from the Paleocene to present, and on what do they base their evidence?
5. Given how few Earth's creatures end up as fossils, do you think we will continue finding new and important fossils, or have we already found most of those that exist?

## 1.9 GLOSSARY



### Glossary

**Biostratigraphy** the use of biological organisms found in rocks to correlate ages between sites.

**Geology**, is the study of the earth, which allows us to understand the preservation, age, and environment in which fossils are found.

**Lithostratigraphy** the use of characteristics of the rock layers to correlate across regions

**Palaentology**, a field that take its name from the Greek words for —oldll (*paleos*) and —existencell (*ontos*), is devoted to gleaning all the information that can be extracted from the fossils.

**Stratigraphy** is the study of the distribution of layers in a soil

**Taphonomy** is the study of what happens to organism from the time they die to their discovery is used. This approach may include both the biological and geological processes.

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## REFERENCES



Behrensmeyer A. Kay & Hill Andrew eds. (1980). *Fossils in the Making*. Chicago: university of Chicago Press.

Brown, Frank, H. (1992). Methods of Dating. In *The Cambridge Encyclopedia of Human Evolution*. S. Jones, R. Martin, D. Pilbeam eds. Pp 179- 186. Cambridge: Cambridge University Press.

Cande S.C. & Kent D.V. (1995). Revised Calibration of the Geomagnetic Polarity Timescale for the Late Cretaceous and Cenozoic. *The Journal of Geophysical Research* 100: 6093- 6095.

Stanford Craig, Allen S. Allen & Anton, Susan C. *Biological Anthropology*. Upper Saddle River, NJ: Pearson Prentice Hall Publishers.

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## THE EARTH IN THE CENOZOIC

### LECTURE TWO

#### 2.1 INTRODUCTION



In this chapter we will undertake to understand the environmental and geographical condition of the earth in the past. Humans are a product of the environment. It determines the kind of economic activities people undertake, where they settle, what kind of plants and animals they exploit. Our understanding of the past environment is therefore of paramount importance since it allows us to understand the conditions that past humans lived in and also allows us to infer the kind of climate they lived in, and also be able to reconstruct the environment.

In order to understand the environmental situation during our period of interest it is important to understand the condition of the landmass during the Cenozoic. Most importantly it is necessary to understand the position of major land masses during this era since it had implications on how animals moved and evolved.

#### 2.2 LECTURE OBJECTIVES



In this lecture students are expected to understand the geological and environmental process that have over the years shaped the earth. It looks analyses position of the continents, reconstruction of the environment and changes over time (different epochs).

#### 2.3 THE EARTH IN THE CENOZOIC



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## 2.3.1 CONTINENTS AND LAND MASSES

The continents have not always been in their current positions. They have shifted over time to their current positions. At around 200 million years ago the earth was divided into two major land masses referred to as Laurasia and Gondwanaland. Laurasia was made up of present-day North America and Asia, and Gondwanaland included Africa and South America. By around 50 million years ago North America and Asia were beginning to spread apart, and South America and Africa had separated from one another and from other continents. Africa eventually became connected to Asia via the Near East, North America and Asia were separated by a chain of Islands, and South America was an island continent until approximately 3.5 million years ago (the Pliocene), when the Central American land bridge connected to North America.

Once the continents were in their present positions, the onset of severe glacial events in the late Pliocene and Pleistocene periodically lowered sea levels, exposing additional land and sometimes resulting, as is the case between continental Asia and Indonesia, in land bridges between otherwise isolated areas.

## 2.3.2 TEMPERATURES

It is important to understand the past environment if we are to deduce the evolution of organisms. Past environments can be reconstructed using several kinds of geological and biological evidence that are referred to as environmental proxies. For our purposes, we are going to use proxies for the reconstruction of temperature, sea levels, and animal and plant communities. The various proxies can be aggregated to from different sites over time and across space to infer how the global environment was like.

## 2.3.3 Oxygen Isotopes, Temperature, and Sea Level

The most used climate proxies are the oxygen- isotope curves that rely on the past ratio of stable oxygen isotope as a proxy for global temperature and sea level. According to Stanford *et al.* (2006) the process works as follows: two stable isotopes of oxygen  $^{16}\text{O}$  and  $^{18}\text{O}$  differ in weight, with  $^{18}\text{O}$  being the heavier among the two. The isotopes exist as oxygen in water molecules and other compounds. In water, they are incorporated in the shells of marine invertebrates that are composed of calcium carbonates. Water molecules formed of the lighter isotope tend to float nearer the ocean surface, whereas water molecules formed of the heavier isotope tends to sink; therefore, the lighter isotope of oxygen tends to evaporate from ocean surfaces sooner than does  $^{18}\text{O}$ . During cold periods when  $^{16}\text{O}$  evaporates from the ocean it is not returned to the world's water reserves via rain but is instead locked up in ice at the poles and northern latitudes. Consequently, sea levels are lower

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during cold periods and contain greater percentage of  $^{18}\text{O}$  than during cold periods. Therefore, the  $^{18}\text{O}/^{16}\text{O}$  ratio increases in sea water during cold periods and in the shells of animals formed in them at the time.

Geologists studying marine cores measure the  $^{18}\text{O}/^{16}\text{O}$  ratio of the marine shells through time to develop the oxygen isotope curves. Higher ratios indicate colder climates and lower sea levels, whereas lower ratios indicate warmer climates and higher sea level. Oxygen isotope ratios can be used to reconstruct local climates by analyzing lake sediments. For example, we can infer the past climate of the East African region by looking at the oxygen ratios from sediments collected from the Lake Victoria.

The Plio- Pleistocene epoch that is important to our study of human origin is characterized by oscillations (intervals) in temperature from colder (glacial) to milder (interglacial) periods. The oxygen isotope stages of glacial and interglacial events are numbered, starting from the most recent. Interglacial periods receive odd numbers because we are in the interglacial period today, which is labeled oxygen isotope stage 1; glacial period receive even numbers. Thus, oxygen isotope curves have been important in correlating important events such as human migrations, but on the finer scale **microclimates** of specific regions are more appropriate.

## 2.3.4 Paleosols and Loess

Soil formation occurs through the process of weathering of the surrounding sediments, and soil structure reflects conditions in the environment at the time that the soil was formed. Using this understanding we can use **paleosol** (ancient soil) to understand the environmental conditions that existed at a given period of interest. From the ancient soil we can also tell the kind of conditions that existed for ancient plants and the plant eating animals. Paleosols can also correlate strata between sites. In addition to paleosol, we can also use windblown sediments (**loess**) in stratigraphy and paleoclimate reconstruction.

## 2.3.5 Vegetation

Plant microfossils or other traces such as root casts may be preserved at some sites. Local plants often are preserved in bog (swamp/ marshland) or peat environments, but plants and plant imprints may also fossilize in very fine- grained sediments. This kind of evidence of the vegetation at paleontological sites can be used to compare the environments that animals once lived in with the present ones. In east Africa Jean Sept used fossilized sediments and vegetation data from ancient riverside habitats to compare the plant diversity in modern east African forests that grow alongside streams with that in the Plio- Pleistocene (Sept 2001)

Recovered pollen can also tell us about the kind of plants that grew in a given site, hence the region.

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Pollen can however be introduced to the site by wind form the surrounding areas. In such a

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situation, the collected pollen is also useful since it can be used to tell the kind of vegetation surrounding the site and in extension the site. Another direct means of assessing plant resources is the presence of **phytoliths** which are opaline silica bodies that are secreted by grasses whose shape is often characteristic of that plant. These bodies can be retrieved through fine, wet screening of sediments. The presence of phytoliths has been used in the analysis of hominid, primate diets and in the interpretation of the uses of stone tools.

## 2.3.6 Stable Carbon Isotope Ratios

Stable carbon isotopes are important in the climatic and environmental reconstruction: they are used to reconstruct the types of vegetation in an area. The science of stable isotope works through the differentiation of the plants using different photosynthetic pathways. Plants use three basic ways to photosynthesize. The most common is the C3 pathways, so called because plants that use it make a three- carbon compound during the first stage of photosynthesis. Trees and shrubs use the C3, as are most of the plants.

The second pathway is known as C4. This is because the plants make a four- carbon compound during photosynthesis. It is believed that these plants use the pathway as an evolution adaptation in response to low levels of carbon dioxide in the environment over the past 50 million years. Almost all tropical grasses are C4 plants, although few grasses and especially the ones from closed canopy environments are C3. All in all, less than 1% of the earth's plants are C4 plants.

The final pathway is composed of plants known as CAM (Crassulacean acid metabolism) whose preventing water loss is critical. CAM plants use C4 pathways at night and as a result of their more complicated pathways tend to lose carbon during photosynthesis. About 4% of the world's plants are CAM plants, including cactus.

Because of their different pathways for photosynthesis, different types of plants retain different amounts of carbon isotopes. C3 plants tend to have less  $^{13}\text{C}$  and hence lower  $^{13}\text{C}/^{12}\text{C}$  ratios. C4 plants have more  $^{13}\text{C}$ . By evaluating those ratios we can tell whether there was abundance of C3, C4, or CAM plants. By looking at these ratios of stable carbon isotopes in the teeth of fossil animals, it is possible to tell the type of vegetation the animal ate and the type of vegetation present in the area. In extension, by looking at the ratios in various animal taxa in an area or through time, it is possible to build what is referred to as a vegetation map. It is also possible to look at stable carbon isotope ratios in the paleosols because the organic carbon found in soils comes from the local plants. This approach has been successful in reconstructing the environments in Africa during hominid evolution particularly because the abundance of C3 and C4 grasses varies between open (savanna) environments and shaded or wooded environments.

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## 2.3.7 Animal Communities

Although some animals can live in almost all kinds of environments, some are very specific to particular habitats. It is common knowledge that hippos and crocodiles live in or near water. The same applies to monkeys who inhabit wooded areas. Further, those animals that are adapted to running long distances over open terrain tend to have longer, slither limbs, while those adapted to life in forested habitats tend to have shorter limbs. Thus, based on comparisons in living animals of known habitat preference we can infer the climatic and environmental preferences of past animals associated with hominid sites.

The animal community approach focuses on all the animals found at particular site during a particular time period, and not just on a single or few species. This is important because the relative abundance of animals can tell more about the environment than few species. However, due to the possibility of fauna being transported by either environmental agents such as water or even animals it is important to exhaustively assess the taphonomic processes that might have played a part in the formation process. Combining the information derived from the communities of animals gives a good impression of the kind of habitat (trees, water etc) and food (grasses, leaves) that were available.

## 2.4 OVERVIEW OF CLIMATIC CHANGES DURING THE CENOZOIC

Based on the different approaches discussed above, scientists have been able to draw a general picture of the climate during the evolution of primates. Thus the section will provide a climatic overview of the Cenozoic and how various changes affected the evolution of human and non human primates.

### 2.4.1 Paleocene to Miocene Climate

The Cenozoic began much warmer than it is today. The temperature differences between the equator and the Poles- North and South was not as distinctively different like it is today. Thus, when primates first arose, they were as equatorial animals as they are today and they existed in abundance further north and south than they do today.

The Paleocene and early Eocene were the warmest of the era. During the Eocene there was a precipitous drop in global temperatures until around the Eocene/ Oligocene boundary (~ 36 million years ago), there was a extreme cold occurrence that led to large extinction and replacement of many species. Such a large die-off is referred to as a —turnover. The temperature changes may have been caused by movement of continents and the resulting changes in ocean and wind currents that alter climate patterns.

From the end of the Oligocene to about the middle of the Miocene temperature rose again

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gradually, although it remained below the levels of Eocene. About the middle of the Miocene, about

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15 million years ago, another abrupt cooling and drying trend happened, which drove the temperatures way below any previously experienced in the Cenozoic. This severe cooling off probably was associated with the appearance of a permanent Antarctic ice sheet at the South Pole. As a result of the evolution of the Antarctic, much water was removed from the oceans, resulting to what is referred to as the Messinian crisis in which the Mediterranean dried up leaving a great salt lake.

## 2.4.2 Pliocene to Early Pleistocene Climate

During the early Pliocene temperatures increased and the Mediterranean Sea refilled, but shortly after, cooling began again. Ice sheets at both [poles became permanent features throughout the Pliocene. The intense cold weather continued to about 2.5 million years ago although there were brief cyclic glaciations in between. According to various researches in the East African region (Kenya and Ethiopia), the period between 4 and 2 million years ago saw changes in climate in form of pulses. These changes show that ungulates went through changes that had major changes in the community of large mammals (Vrba 1996).

The changes in animal species in the region show some relation to the appearance of the genus *Homo*. Later periods of climatic changes was linked to the appearance of early stone tools. Later fluctuations were linked to the apparent extinction of some australopithecine taxa.

## 2.4.3 Early to Late Pleistocene Climate

At around 2.5 million years the glacial cycles became more severe. The glaciations were so severe that it lowered sea levels enough to connect island southeast Asia to mainland Asia. This connection was important as hominids started to move out of Africa. An important aspect of increasing cold and glacial cycling is that latitudinal variation in climate became quite significant. As the hominids began to move out of Africa around 1.8 million years into western Asia and into northern altitudes of Europe (~ 800,000 mya) climatic conditions in some instances were quite harsh. This appears to have been a hindrance to the hominids moving further north until much later with the help of mitigating cultural adaptation such as fire, clothing and shelter necessary to fight the intolerable elements.

## 2.5 SUMMARY



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Understanding the past environment is important in the reconstruction of the past. Understanding the past environment becomes a central focus if we are to understand the past. There exist various ways both biological and geological that aid us in the inferring the past environments.

## 2.6 SELF- TEST QUESTIONS



1. According to your understanding how are any of the following environmental proxies used in the reconstruction of past environments?
  - a) Oxygen isotopes
  - b) Animal communities
  - c) Stable carbon isotope ratios
  - d) Paleosols and loess
  - e) Vegetation
2. What is the general believe by scientists on climate change from the Paleocene to the present? What do they base their evidence on?
3. Why is the stable carbon isotope analysis approach ideal for the reconstruction of the East African paleo- environments?

## 2.7 GLOSSARY



**Loess** windblown sediments

**Paleosol** ancient soil

**Phytoliths** opaline silica bodies that are secreted by grasses whose shape is often characteristic of that plant.

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## REFERENCES



- Alvarez L.W, Alvarez W, Asaro F, Mitchel H.V. (1980). Extraterrestrial cause for the Cretaceous-tertiary extinction. *Science* 208: 1095- 1108.
- Vrba, E.S. 1985. Ecological and adaptive changes associated with early hominid evolution. In *Ancestors: the hard evidence* (ed. E. Delson), pp. 63-71. New York, Alan R. Liss.
- Ward, C.V. (1997). Functional anatomy and phylogenetic implications of the hominoid trunk and hindlimb. In *function, phylogeny and fossils: Miocene hominoid evolution and adaptation*. DR Begun, CV Ward, MD Rose Eds. Pp 101- 130. New York: Plenum Publishers.

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## ORIGINS OF PRIMATES

### LECTURE THREE

#### 3.1 INTRODUCTION



In this chapter, we will look at the origin of primates. Our understanding of the origin is important since it will give us a background of the origin of mammals and primates which will be helpful in the understanding the concept of human evolution. The theory of human evolution which will be dealt with in the upcoming chapters postulates that modern humans are related to other mammals and especially primates. Thus the understanding of origin of primates becomes important if we are to understand the history of human origin.

Our review of primates starts at the Mesozoic Era ; 225- 65 Mya because the origin of primates is tied to the origin of mammals. During this era, critical environmental changes occurred that provided opportunities for small, insect eating- eating mammals from which primates evolved.

#### 3.2 LECTURE OBJECTIVES



In this lecture students are expected to understand the process, dates and conditions of the evolution of mammals and primates. This is important because an understanding of the above gives the student a background to the evolution of humans and conditions necessary to the evolution of the same.

#### 3.3 ORIGINS OF MAMMALS



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The mammals in the Mesozoic were small creatures greatly overshadowed by the dinosaurs. According to Stanford *et al*, the mammals that first appeared in the late Triassic Period most closely resembled the living monotremes, such as the echidna and platypus. In the Jurassic Period the first marsupial and placental mammals appeared and both diversified greatly in the final period of the Mesozoic, the Cretaceous.

## 3.3. ORIGINS OF PRIMATES

### The K-T Boundary

At the end of the Mesozoic, drastic environmental changes most likely caused by an asteroid or satellite crashing into the surface of the earth, caused or contributed to the extinction of the dinosaurs and generated opportunities for mammals (Alvarez et al. , 1980). The evidence of such an occurrence is the presence of a giant crater called Chicxulub in the Yucatan Peninsula.

The impact probably caused an all consuming firestorm and a number of tidal waves. This was followed by abrupt global cooling. It is thought that this combination of fire and cold consumed much of the land plant life at the time causing the extinction of herbivorous dinosaurs and then also of the carnivorous dinosaurs that preyed on them.

The ensuing climatic and ecological conditions favored small insect- eating mammals that with the absence of carnivorous large dinosaurs thrived. Some of the primitive mammals of the Mesozoic persisted into Paleocene, the earliest Cenozoic epoch, but for the most part there is a comprehensive replacement of mammals at the K-T boundary. Many of these mammals including the ancestors of primates are of archaic forms that are not traceable to living groups.

### 3.3.1 Changes In The Paleocene

During the Pliocene Epoch, many archaic groups of mammals arose that are not similar to many of the living groups. Among these groups it includes the ancestors of the living mammalian orders including the primates. Among these early forms is the **plesiadapiforms**, which is thought to be an ancestor of primates although much controversy about this exists.

Paleontologists use the form of teeth and bones to decide to which groups a fossil belongs to. Thus, for a fossil to be identified as primates, fossils must show the primate trends in anatomy. The trend in anatomy is the basis of controversy around the **plesiadaforms**, which in many ways is more primitive than primates. According to Stanford et al.(2006), they had small brains, a **prognathic face** that projected well in front of their brain case, small eye sockets positioned on the sides instead of the front of their faces. They lacked a **postorbital bar**, a bony ring encircling the eye, a key feature that characterizes primates, indicating the importance of vision to the order. Many plesiadaforms possessed large, rodent like lower incisors that were separated from the premolars by

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a large diastema, a gap between their anterior teeth. Some had claws rather than nails and lacked an opposable big toe. In all these ways, plesiadapiforms do not look like primates.

From ancestors such as the plesiadapiforms, an ***adaptive radiation*** of plesiadapiforms evolved, ranging from the very tiny, and mouse- sized (about 20gms) forms to creatures that are about the size of a small monkey (5 kgs). These include the genus *Plesiadapis*, a rodent- like animal from the early Paleocene and Eocene of Europe and North America that seem to have moved slowly along tree branches and perhaps traveled on the ground as well, and the paromomyidae, a family of plesiadapiforms that lived from the middle Paleocene until the late Eocene of Europe and North America, that may have fed on sap and gum that flowed from places that these animals had gouged in the tree bark.

In the Paleocene the climate was warmer than today, and was a period of recovery from the giant impact in K-T boundary. In this Epoch, flowering plants evolved, insects increased in number and diversity as pollinators for these plants, and the plants evolved visual cues to lure insects. Primate ancestors cashed on the available insects and also possibly the fruits that came with the new plants.

## 3.4 EARLY PRIMATES OF THE EOCENE

Climate warmed significantly in the beginning of the Eocene, around 54 million years ago, resulting in the replacement of the archaic mammals of the Paleocene by the first representatives of a number of modern orders of placentals including Rodentia (mice, squirrels etc), Artiodactyla (even-toed ungulates or hoofed animals such as deer, camels, hippos, and pigs), Perissodactyla (odd- toed ungulates such as horses and rhinos) and primates.

The fossil record of the Eocene reveals the first true primates, those that possess the bony characters by which we identify primates with. The two main super families of Eocene primates, the Adapoidea and Omomyoidea, appeared at the beginning of the Eocene. They flourished during the Eocene of Europe, North America, Middle East, Asia, and North Africa but declined in the Oligocene.

These fossils are accepted as true primates because unlike the earlier plesiadapiforms, they possess the suite of primate characteristics. In particular, they possess slightly larger brains than plesiadapiforms, eye sockets positioned on the front of the face which allows stereoscopic vision, a complete postorbital bar for protection of the eye, an opposable big toe, and nails at the end of their digits. At the same time, reduction in their snouts suggests that smell was not as important in locating food as vision.

### 3.4.1 ADAPOIDS (STREPSIRHINE ANCESTORS)

The adapoids resemble modern strepsirhines mainly in primitive ways, and most lack the features of modern primates such as the tooth comb. The adapoids are best considered the most primitive known group of early modern primates that probably gave rise to the strepsirhines. The adapoids

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were mostly small-to- medium and weighed approximately 100g to 7 kgs. They were slow- moving tree dwelling quadrupeds that were active by day and probably lived on a diet of fruits and leaves.

The most successful superfamily **Adapoidea** is divided into three families: the Northarctidae, Adapidae, and Sivalapidae (Fleagle, 1998). They are mostly abundant in the Old World. Fossils show long broad snouts with teeth that suggest some may have eaten fibrous diet. Their eyes indicate that some were probable nocturnal, others diurnal. Their postcranial indicate a diverse range of locomotion from leaping to quadrupedal climbing. Although some adapoids are likely to be ancestral to living strepsirhines (lemurs and lorises), the true fossil record of the true lemurs is confined to the Holocene of Madagascar.

## 3.4.2 OMOMYIDS (HAPLORHINE ANCESTORS)

**Omomyoids** are Eocene primates that had recently diverged from the adapoids and may have given rise to the common ancestor both tarsiers and anthropoids. The omomyoids were smaller-bodied primates (30g -5kgs) that ate diets of insects and fruit and had larger orbits, probably for nocturnal lifestyle. Their structural evolution was probably set up for active arboreal quadrupedalism and leaping like those of living galagos. Omomyoids are more abundant in North America although there is evidence of their presence in Europe too.

## 3.5 CONTINENTAL DRIFT AND EOCENE PRIMATES

The geographic distribution of the adapoids and omomyoids in North America and Europe is understandable considering the positioning of the two continents between 54 and 34 million years ago. At this point, Europe and North America were joined by a broad band of land, and there was minimal difference in climate from north to south or east and west.

### 3.5.1 EVOLUTION OF HIGHER PRIMATES

Representatives of the higher primates i.e. monkeys and apes first appeared in the late Eocene and early Oligocene epochs after the strepsirhines – haplorhine split. The earliest higher primates are generalized monkeys that probably gave rise to all later higher primates. Early apes appeared in the Miocene and were more generalized than their living descendants and more diverse. The initially diverse apes decreased in abundance through time, although monkeys became more diverse.

### 3.5.2 THE FIRST MONKEYS

**The Grand Coupure** which was a large- scale faunal turnover at the end of Eocene created challenges and opportunities for animal populations alive at the time. As a result, the adapoids and omomyoids nearly vanished from North America and Europe, same as many other mammalian taxas. Molecular evidence shows that the first monkeys occurred between 58 and 40 million years ago, after the split between tarsiers and anthropoids occurred, and before New World and Old World diverged.

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The fossil record provides peeks into the origin of anthropoids. One is in the Eocene of China, the other in North Africa and the Middle East. Much of what is known about the evolution of higher primates comes from research done at the Fayum depression in Egypt. This region during the Eocene and Oligocene it was a lushly forested area surrounding a large river system that supported a great diversity of tropical flora and fauna. Few several genera of small (anything between 500-1000g) early anthropoids in the Fayum deposits have been recovered. These forms only form fragmentary jaws, teeth, and isolated postcranial elements. These early forms from Fayum combine primitive features and anthropoid- like features.

It is not until the end of the Eocene and the early Oligocene that there is clear evidence of anthropoids in the fossil record. There are three families of these early anthropoids, all from Fayum depression: the Parapithecidae, the Oligopithecidae, and the Propliopithecidae. Like modern higher primates, these early anthropoids possessed advanced features of the skull and jaws, including a fused frontal bone, a fused lower jaw, and postorbital closure, that distinguish them from strepsirhines (Stanford *et al.* 2006)

### 3.5.3 NEW WORLD MONKEYS

New world monkeys are monkeys in the Americas. It is important to understand their relationship with the Old world primates since it would explain the evolution of primates. The molecular evidence suggests that New World and Old World monkeys diverged around 40 million years ago. The history of how the primates got to South America is a mystery, since South America was an island continent during the early part of the Cenozoic.

What is however more controversial is the question of whether the New World monkeys originated from an advanced stock of Eocene strepsirhines in North America or from the most primitive anthropoids of Africa. Perhaps New World monkeys descended from North American Eocene primates that migrated south across open water ways between islands. New World monkeys would therefore have evolved in parallel with Old World monkeys and would be only distantly related to them. Alternatively New World monkeys have three premolars rather than the two seen in all catarrhines, linking them to Apidium, a fossil monkey from Egypt. In the late Eocene and early Oligocene, the Atlantic Ocean would have been far less of a geographic barrier to dispersal from Old World because it was not as wide as it is today. In addition, the molecular evidence shows that the two groups split recently, about 40 million years ago. Because of the molecular and anatomical evidence, most scientists support a model that supposes an Apidium- like ancestor somehow ‘floating over’ from Africa to South America during the late Eocene or early Oligocene.

### 3.5.4 OLD WORLD MONKEYS

Going by molecular evidence, Old World monkeys and apes shared a common ancestor about 25 million years ago. The fossil records shows that the ancestor had anatomical features shared by both monkeys and apes, such as a bony ear tube and presence of two rather than three premolars but lacked characters unique to each group, such as bilophodont molars characteristic of the modern

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Old World monkeys and the suspensory shoulder characteristic of the modern apes.

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The earliest fossil evidence of a lineage leading to Old World monkeys comes from a 19 million year old site of Napak in Uganda. The monkey represents an early radiation of the family Victoriapithecidae that predates the split between the subfamilies of the modern leaf-eating Colobinae and fruit-eating Cercopithecinae. The specimen lived in the early to middle Miocene of eastern and northern Africa, but they are best known from thousands of specimens of the species *Victoriapithecus macinessi* from 15- million years- old deposits at Maboko Island (Benefit and McCrossin 2002).

*Victoriapithecus* was a vervet- sized monkey weighing between 3 and 5 kg. Its limb remains indicate that it used a semi terrestrial pattern of locomotion, making it one of the oldest and smallest anthropoid primates to make the shift to life on the ground. Its low molar cusps and broad upper incisors indicate that it probably ate hard fruits and seeds. The Miocene monkeys disappeared from the fossil record approximately 12 million years ago to be replaced by true Colobines and Cercopithecines.

The molecular evidence suggests that within the Old World monkeys, the cercopithecine and colobine split dates to between 14 and 16 million years ago. The first true cercopithecine (*Macaca* sp.) appears on the fossil record around 11 million years ago in North Africa. In Eurasia, cercopithecines are represented by numerous Pleistocene Asian macaques. The first true colobines appear in Kenya at around the same time period (11 million years go). Fossil record shows that soon after appearing in Africa, colobine monkeys expanded their range to Eurasia.

## 3.5.5 THE EARLIEST APES

The presence of living ape fossil is limited to four genera: *Hylobates* which represents the gibbons and Siamangs, *Pongo* (orangutan), *Gorilla* (the gorilla), and *Pan* (the bonobo and common chimpanzee). However, the fossil record of hominoid primates reveals a diverse succession of adaptive radiations. This ape fossil record is characterized first by the appearance of dental apes, animals with apelike teeth but monkey- like postcranial skeletons. Fossil apes first appeared during the Early Miocene, approximately 23 to 16 million years ago. At this time, fossil record shows that the hominoids were predominantly restricted to Africa with sites in Kenya, Uganda, Tanzania and Namibia. The early Miocene in Africa was covered by uninterrupted expanses of forest and moist woodland.

The dental apes were small- bodied compared with modern apes, they lacked a suspensory shoulder for brachiating, and they walked in plantigrade fashion i.e. on the soles of their feet rather than on their knuckles. The best known of the dental apes is the genus *Proconsul* which lived in Africa about 18 to twenty million years ago. Their teeth are apelike, with aspects of the limb skeleton showing a monkey- like locomotor adaptation for running. *Proconsul* appears to have possessed a long and flexible torso, like that of monkeys rather than the short and stable back of living suspensory apes and a possibility of a tail (Ward, 1997).

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## 3.6 SUMMARY



As shown in this chapter, in order to understand human evolution and adaptation, it is important to have an understanding of the history of the origin of mammals and primates. Primates are especially important since they are the most closely related mammals to humans. Evolution of apes in Africa is also important in the understanding of human evolution in the continent in the Miocene and Pliocene. The chapter also brought out the important role that the African continent has played in human evolution and adaptation.

## 3.7 SELF- TEST QUESTIONS



1. How are adapoids and omomyoids similar, and how do they differ?
2. Why the earliest apes are not easily identified as such?
3. How would the rise of rodents in the Eocene have contributed to the demise of the plesiadapids? What evolutionary principles can be applied here?

## 3.8 GLOSSARY



**Plesiadapiforms**, primitive form in the Pliocene thought to be an ancestor of primates although much controversy exist.

**Postorbital bar**, a bony ring encircling the eye, a key feature that characterizes primates, indicating the importance of vision to the order.

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## REFERENCES



Alvarez L.W, Alvarez W, Asaro F, Mitchel H.V. (1980). Extraterrestrial cause for the Cretaceous-tertiary extinction. *Science* 208: 1095- 1108.

Ward, C.V. (1997). Functional anatomy and phylogenetic implications of the hominoid trunk and hindlimb. In *function, phylogeny and fossils: Miocene hominoid evolution and adaptation*. DR Begun, CV Ward, MD Rose Eds. Pp 101- 130. New York: Plenum Publishers.

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## THE BASIS FOR HOMINID EVOLUTION

### LECTURE FOUR

#### 4.1 INTRODUCTION



In this chapter we will look at some factors that had to be present for human evolution process to have occurred. Some factors such as origin of bipedalism and the presence of species that were to undergo changes that led to modern humans. Bipedalism which is the ability to walk on two feet was necessary because it allowed those species to concur the landscape easily, freed their hands for other activities e.t.c.

#### 4.2 LECTURE OBJECTIVES



In this lecture students are expected to understand the process of human evolution from the primitive hominid species. It will also analyze conditions and factors such as bipedalism and presence of various species that eventually played a significant role in the evolution of modernity in humans.

#### 4.3 THE BASIS FOR HOMINID EVOLUTION



##### 4.3.1 BIPEDALISM

*Bipedalism* refers to locomotion (e.g., walking, jogging, running, etc.) on two legs. It is not uncommon to see animals standing or walking on two legs, but only a few animals practice bipedalism as a consistent means of locomotion. Most other animals, including chimpanzees and gorillas, practice a form of locomotion called *facultative bipedalism*, a type of bipedalism assumed on a

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temporary basis in order to perform a particular function.

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The *striding gait* of human bipedalism involves the fluid flow of a series of actions collectively, the swing phase and the stand phase. One leg alternating with the other. The leg in the swing phase pushes off with the power of the toe, swing under the body in a slightly flexed position. Finally become extended as the foot make contact with the ground first with the heel. Once heel strike occurred, the leg remained extended providing support for the body. The stance phase, while the other leg goes through the swing phase with the body moving forward.

Two key features differentiate human and chimpanzee bipedalism:

- Chimpanzees are unable to extend their knee-joints to produce a straight leg in the stance phase.
- Muscular power has to be exerted to support the body.
- The constantly flexed position of the chimpanzee leg also mean there is no toe off and heel strike in the swing phase.

## 4.3.1.1 Possible Theories for the Origin of Bipedalism

### A. Locomotor efficiency

Advocated by two researchers —Peter Rodman and Henry McHenry, who examined the locomotor efficiency of bipedalism. They found bipedalism and quadrupedalism to be similar in energy costs; however, an examination of our closest living relatives (the great apes) suggests that perhaps early hominids adopted a similar suspensory posture, from which the move to bipedalism could be facilitated.

### B. Thermoregulation

Peter Wheeler suggested that an erect posture was adopted by early hominids to combat high temperatures, especially when traveling exposed to direct sunlight (as in the open savanna). The benefits of bipedalism are twofold: the reduction of exposed area of the body to sunlight and an increased exposure to cooling winds.

### C. Freeing hands for other uses

The benefits of being able to transport things manually are obvious. Some species of primates have adapted mechanisms to perform these tasks (for example, the cheek pouches in cercopithecines); however, the ability to carry a resource either to save for later or to consume in a safer location would allow flexibility in planning for early hominids.

### D. Postural feeding

Kevin Hunt recently proposed that the emergence of bipedalism came as a postural feeding

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adaptation. Observing common chimpanzees in the wild, he determined that, although they rarely

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walk bipedally, when they choose to adopt a bipedal position it is almost always in the context of feeding on fruit from small trees. This increases foraging efficiency, particularly when fruiting trees are spread about. Feeding chimpanzees can shuffle over to the next tree, rather than get down on all fours, move to the next tree, and then get back up.

## 4.4 DIVERGENCE OF THE HUMAN LINEAGE FROM OTHER GREAT APES

Species close to the last common ancestor of gorillas, chimpanzees and humans may be represented by *Nakalipithecus* fossils found in Kenya and *Ouranopithecus* found in Greece. Molecular evidence suggests that between 8 and 4 million years ago, first the gorillas, and then the chimpanzees (genus *Pan*) split off from the line leading to the humans; human DNA is approximately 98.4% identical to that of chimpanzees when comparing single nucleotide polymorphisms. The fossil record of gorillas and chimpanzees is quite limited. Both poor preservation (rain forest soils tend to be acidic and dissolve bone) and sampling bias probably contribute to this problem.

Other hominines likely adapted to the drier environments outside the equatorial belt, along with antelopes, hyenas, dogs, pigs, elephants, and horses. The equatorial belt contracted after about 8 million years ago. Fossils of these hominans - the species in the human lineage following divergence from the chimpanzees - are relatively well known.

The earliest are *sahelanthropus tchadensis* (7 Ma) and *Orrorin tugenensis* (6 Ma), followed by: australopithecines.

### 4.4.1 AUSTRALOPITHECINES

The hominid family originated between 10 and 5 million years ago, with a single species of bipedal ape. As usually happens with newly established mammalian lineages: The 1<sup>st</sup> specie gave rise by stages to a range of descendants, producing a relatively luxuriant evolutionary bush. Individual branches were pruned away from time to time as species became extinct and new ones were added. Eventually, the hominid bush was reduced to just *Homo sapiens*.

No less than 3 perhaps 6 and maybe more hominid species coexisted 2 million years ago in Africa. Those species can be divided into 2 groups:

1. With relatively large brains and small cheek teeth.
2. Large-brained species were member of the genus *Homo*.

If several species of *Homo* existed at this time, only one of them could have been ancestral to modern humans, while the rest became extinct. Labeling of the 2<sup>nd</sup> group is more contentious. We will call them australopithecines; they all became extinct.

#### **Anatomy and biology of the australopithecines:**

- Bipedal apes with modified dentition.
- Lived in more open environmental setting, not the open plains of bushland and wooden

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Savannah.

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- Hominid structure of teeth and jaws appear to have required more grinding than an ape's diet.
- Male australopithecines were larger in body size. 20-40% taller, 30-40% heavier than females.
- Australopithecines were social animals.
- Foraging strategies of hominids were not dramatically different: australopithecines have been carnivores.
- Australopithecines were principally vegetarian.

Australopithecines of 2 million years ago occurred in 2 forms:

- a) **Gracile (means slender)**
- b) **Robust.**

So far, only one gracile australopithecine has been identified whereas, as many as 4 robust species have been named. In South Africa, the gracile species is *Australopithecus africanus* and the robust *Australopithecus robustus*. The robust australopithecine in East Africa is *Australopithecus boisei*. Naming gracile species in East Africa is more contentious; some apply the name *Australopithecus africanus* to some specimens. The term gracile and robust implies substantial anatomical differences between the 2 forms. One small and delicately built. The other bigger and more massive. Scholars realize the difference between the 2 forms is mainly in the dental and facial adaptations to chewing. The robust forms have bigger grinding teeth, more robust jaws and more bulky chewing muscles and muscle attachments.

The remainder of the skeleton: The gracile and robust australopithecines were roughly comparable, with the robust having a slightly greater stature. Brain size, based on small specimens, gives the robust an edge over the gracile. Both are close to 500 cm<sup>3</sup>. Most scholars agreed hominids did not leave Africa before about 1 million years ago, when *Homo erectus* expanded to Eurasia.

## **Australopithecines anatomy**

- Teeth, jaw and cranial anatomy are one functional complex.
- Difference between the 2 forms of australopithecine is that the robust species have taken this adaptation to an extreme, having enormous, flat molars and relatively small blade-like incisors and canines.
- In all hominid, the tooth row is tucked under the face more than the apes, giving a less projecting facial profile and increasing chewing efficiency.
- The robusticity of the lower jaw (mandible) that is characteristic of hominids compared with apes is particularly apparent in the robust species, reflecting more powerful chewing action.

Extra muscle in the robust has 2 anatomical consequences:

- a) One of the muscles that power the lower jaw-the temporal muscle-is anchored to a raised bony crest that runs along the top of the cranium, front to back.
- b) This so-called sagittal crest also in gorillas is absent in gracile australopithecines.
- c) The great size of the temporal in robust and a second chewing muscle, the **masseter**, causes the cheekbones (the zygomatic arch) to be exaggerated and flared forward.

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This and the strengthening of the central part of the face by pillars of bone, gives the robust australopithecine face a characteristic dished appearance. The difference in dental apparatus between gracile and robust australopithecines was once interpreted as a result of substantial differences in diet. Robust was more vegetarian. Gracile more carnivorous. The past decade they were implies a vegetarian diet for the australopithecines. More recent Fredrick Grine and Richard Kay concluded robust consume tougher foods than gracile. It is consistent with evidence that robust australopithecine's lived in drier habitats where soft fruits and leaves were more common. Sillien recently challenged this conclusion. His analysis included the robust species included a significant amount of meat.

Function and overall sizes:

- Post cranial skeletons (neck down) of gracile and robust australopithecines are similar to each other.
- *Australopithecus africanus* weights 41/30 kilograms.
- *Australopithecus robustus*, 32/40 kilograms.
- *Australopithecus boisei*, 49/34 kilograms.

The australopithecine pelvis of 2 million years ago was like Lucy a million years earlier. The thighbone is different from Homo pattern. Head of the femur is smaller than in Homo and attached to a longer, slender neck. Individual bones indicate although these species were bipedal, they were as adapted to climbing as was *Australopithecus afarensis*. There were differences between Lucy's hands than later australopithecines. The handbones of *Australopithecus afarensis* were apelike-having curved phalanges, thin tips to the finger and a short thumb. Recent analysis of robust australopithecine hand bones indicates they were much more humanlike. The thumb is longer and more mobile, fingertips were broader. The robust australopithecines probably possessed sufficient skills to make stone tools. These are differences of opinion. Simple bone tools-digging sticks have been recovered from were *Australopithecines robustus* fossils are known. The possibility remains they were made by Homo. The robust and gracile australopithecines were viewed as same animal but built on different scales.

## 4.5 SUMMARY



The process of human modernity was greatly influenced by biological and environmental factors that had to be present. The presence of apes and the evolutionary processes that were at play made the transition from apes to hominid- like species possible. Other biological factors such as the ability

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to walk on two legs also played an important role in the success of the evolution to modernity in humans.

## 3.7 SELF- TEST QUESTIONS



1. What are the possible explanations as to the origin of bipedalism?
2. What are the possible explanations as to the minimal fossil evidence for apes?
3. Why are gracile and robust australopithecines considered same species
4. How can we explain the anatomical differences between gracile and robust australopithecines?

## 3.8 GLOSSARY



**Bipedalism** refers to locomotion e.g., walking, jogging, running, etc., on two legs.

**Gracile** means slender. Most of them were relatively small, slender, and delicate boned

**Robust** larger jaws accompanied by pronounced sagittal crests in the case of males. They also had much larger back

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## REFERENCES



Dart, R.A. 1925. *Australopithecus africanus*: The man-ape of South Africa. *Nature*, 115, 195-199.

Dart, R.A. 1960. The bone tool-manufacturing ability of *Australopithecus prometheus*. *American Anthropologist*, 62, 134-143.

Domínguez-Rodrigo, M., J.A. López-Saez, A. Vincens, L. Alcalá, L. Luque and J. Serrallonga. 2001. Fossil pollen from the Upper Humbu Formation of Peninj (Tanzania): hominid adaptation to a dry open Plio-Pleistocene savanna environment. *Journal of Human Evolution*, 40, 151-157.

Fleagle, J.G. Primate adaptation and evolution, 2nd ed. Academic Press, San Diego.

Kimbel, W.H., Y. Rak and D.C. Johanson. 2004. The skull of *Australopithecus afarensis*. Oxford, Oxford University Press.

McHenry, H.M. and L.R. Berger. 1998. Body proportions in *Australopithecus afarensis* and *A. africanus* and the origins of the genus *Homo*. *Journal of Human Evolution*, 35, 1-22.

Stanford, C. 2003. Upright: The evolutionary key to becoming human. Houghton Mifflin, New York.

Tobias, P.V. 1965. Early man in east Africa. *Science*, 149, 22-33.

Tobias, P.V. 1998. Ape-like *Australopithecus* after seventy years: was it a hominid? *The Journal of the Royal Anthropological Institute*, 4, 283-308.

Vrba, E.S. 1985. Ecological and adaptive changes associated with early hominid evolution. In *Ancestors: the hard evidence* (ed. E. Delson), pp. 63-71. New York, Alan R. Liss.

Walker, A., R.E. Leakey, J.M. Harris and F.M. Brown. 1986. 2.5-Myr *Australopithecus boisei* from west of Lake Turkana. *Nature*, 322, 517-522.

Walker, A. and P. Shipman. 1996. The wisdom of bones: in search of human origins. New York, Vintage Books.

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## ORIGINS OF THE GENUS *HOMO*

### LECTURE FIVE

#### 5.1 INTRODUCTION



The origin of the genus *Homo* is important to our understanding of origin of modernity since it marks the actual evidence of modernity in humans as we know it. However our understanding of the when, where and whom modernity began with is shrouded in mystery. Over the last decades fossil records have sparked new, revolutionary ideas about the origins of the genus *Homo*. Of importance is whether the first truly human primates evolved in East Africa, as the textbooks would have it? Or was the human homeland in Asia or South Africa? This chapter will unravel the mystery of origin of modern humans as the evidence shows.

#### 5.2 LECTURE OBJECTIVES



In this lecture students are expected to understand the process of human evolution from the primitive hominid species. It will also analyze conditions and factors such as bipedalism and presence of various species that eventually played a significant role in the evolution of modernity in humans.

#### 5.3 GENUS *HOMO*



The evolution of the modern human genus can be divided roughly into three periods: early, middle, and late. Species of early *Homo* resembled the early australopithecines in many ways. Some early *Homo* species lived until possibly 1.6 million years ago. The period of middle *Homo* began perhaps

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between 1.8 million and 2.0 million years ago, overlapping with the end of early Homo. Species of

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middle Homo evolved an anatomy much more similar to that of modern humans but had comparatively small brains. The transition from middle to late Homo evolved large and complex brains and eventually language. Culture also became an increasingly important part of human life during the most recent period of evolution.

The key change usually considered to signal the origin of Homo is an increase in brain size, measured by the volume of the inside of the brain case (cranial capacity). The average cranial capacity of modern humans (*Homo sapiens*) is 1350 milliliters (about 5 cups), although the range of variation is large, around 1000 to 2000 cc. In the possible ancestors of Homo (*Australopithecus afarensis* and *A. africanus*) brain size was about 350 to 500 ml. What size, it may be asked, defines the difference between the brains of Homo and Australopithecus?

The first discoveries of the genus were made at Olduvai Gorge. A series of anatomical characters is to be found uniquely in *Homo*, such as an increase in cranial vault, reduced lower facial prognathism, reduction in the size of premolars and molars, and the length of the molar row.

## 5.3.1 *Homo erectus*

At some point around the Plio- Pleistocene boundary at around 1.8 million years ago, hominids underwent a major adaptive shift. This shift is evidenced in the fossil record where a number of changes such as lengthening of the femur which indicates a change in the gait and general body size increase to almost that of modern humans. These changes might have been as an adaptation to environmental changes at the time.

*Homo erectus* appears in Africa at around 1.8 Mya, and evidence shows that it was the first species to leave Africa at around 1.7 Mya. Some of the oldest *Homo erectus* sites outside the continent are Dmanisi in Georgia and Sangiran in Indonesia. These two sites have versions of the species, although at Dmanisi the specimens are slightly smaller compared to the typical *Homo erectus* though quite similar to the small- brained early versions of the early *H. erectus*. The species existed up to around 300 Kya, in various places at the same time. It is evident that in some places the species had already evolved into other new forms or in the process of doing so.

## 5.3.2 ANATOMICAL FEATURES

**Skull and Teeth** the skull is thick- boned and robust, much longer than it is wide, relatively low and angular from the side. The angular features of the skull are enhanced by cranial superstructures, regional thickening of the bone along sutures and across certain bones. These include thickenings such as the prominent *supraorbital torus* or the brow ridge on the frontal, a thickened *angular torus* on the back of the parietal, and the *occipital torus*, a ridge of bone that runs horizontally across the occipital (Fig. 3.1). The forehead also has a low, sloping or receding appearance that is separated from the supraorbital torus by a furrow. The pentagonal rear view is formed by the thickening including those along sutures such as the sagittal *keel* along the sagittal suture that joins the two parietals and the *metopic keel* along the midline frontal.

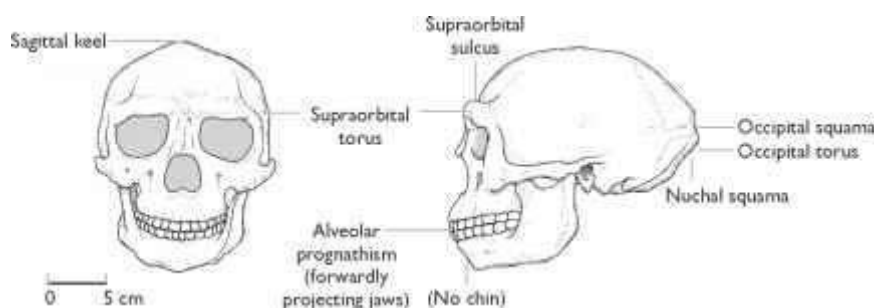
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*H. erectus* brain size ranges from approximately 700 cc to 1,200 cc, averaging about 900 cc (Stanford *et. Al* 2006). The brain size increase in *H. erectus* may have been an increase in proportionality to body size, but could also be as a result of increased cognitive abilities. The brain sizes show regional and evolutionary variation, indicating progressive but slow increase in the regional lineage through time (Anton and Swisher, 2004).

The jaw of the species was robust and powerfully built as the rest of the crania. It was probably so, as a result of the diet that the species consumed. The premolars and molars also display a complex root system, with the incisors being concave in shape, with ridges along their adges forming a *shovel-shaped incisor*.

**Body Size and Shape** although there have been many cranial fossil of the species found, there are very few complete skeletons of the species on record. The few post cranial finds are mainly from East Africa with the most complete and prominent find being KNM-WT 15,000 otherwise referred to as Turkana boy or Nariokotome boy found in 1984 on the western side of Lake Turkana by Alan Walker and Richard Leakey.



**Fig. 3.1: Cranial features characteristic of *H. erectus***

The specimens suggest that *H. erectus* was robustly proportioned and that some of the individuals were quite tall as adults. They according to Walker (1993) could grow to between five feet and six feet Based on KNM- WT 15,000. Their lower limbs were long, with a narrow hip. These body proportions (long and linear) seem to follow human adaptation to tropical environments, suggesting that the species was dissipating heat like modern humans by sweating. This suggests that the species was quite active during the day, even the hottest times- noonish, since they would dissipate heat.

## **5.3.2 HOMO ERECTUS AROUND THE WORLD.**

*H. erectus* as a species spanned quite a substantial time. It represents more than 1.5 million years of time and a broad geographic range (Fig. 3.2). Sites so far range from about 1.8 million years to about 100,000 years. In Africa where *H. erectus* is first found, it persisted from 1.8 million years ago to about 1 million years ago; in Georgia to about 1.7 million years ago; in island S.E. Asia by about 1.8

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million years and in mainland Asia up to about 200,000 years ago. These geographical ranges can explain various evolutionary differences especially anatomically between various specimens of the species.

The earliest fossil record for *H. erectus* comes from Koobi Fora at 1.8 million years ago. The oldest remains are a largely complete cranium KNM-ER 3733, dated at 1.78 million years old, with a cranial capacity of about 850 cc. there have been other older specimens of about 1.9 million years old but are too fragmentary or their identification suspect. The western side of the lake has also produced *H. erectus* species with the most prominent being the Nariokotome boy with an estimated age of between 8 and 11 years given.

Other important *H. erectus* sites include the Olduvai Gorge where the largest brained specimen of the species in Africa, OH 9, with a cranial capacity of slightly more than 1,000cc dating to about 1.47 million years old was found. The site also has some of the smallest brained *erectuses* with OH 12 with a capacity of 727 cc and dated to about 780,000 years ago. Olorgesailie also recently produced a *H. erectus* dated to around 900,000 years ago (Potts 2004). In Ethiopia, Bouri Formation of the Middle Awash and Danakil depression in Eritrea has produced *H. erectus* dated to about 1 million years old.



**Fig. 3.2. *Homo erectus* around the world.**

Outside Africa, the most prominent *H. erectus* site is at Dmanisi in the republic of Georgia. They were first recovered in the 1990's in an abandoned well beneath a village. Four crania and post crania remains of *H. erectus* fossils have so far been found dating to about 1.7 million years ago. These hominids are very similar to early African *H. erectus* but with relatively smaller brains, on average, less than 800 cc. They are according to Stanford *et al*( 2006), linked to the African species by their premolar and molar tooth structure, the development of the brow ridges and the high cranial vault. The Dmanisi find thus shows that early humans had migrated out of Africa at nearly the same time that the species appears in Africa.

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In Asia, the oldest species of *H. erectus* is from the island of Java in South East Asia and dated to about 1.6 – 1.8 million years old. It is important to note that at 1.8 million years ago, the sea level in south East Asia was relatively low than it is today and the Island of Java was part of the mainland. In mainland Asia the first *H. erectus* fossil dates to around 1.5 million years ago. It was collected in Indonesia in 1891 by Eugene Dubois in the banks of Solo River at the village of Trinil in Indonesia. The species have also been found in China. The species range between 800,000 and 200,000 years ago.

## 5.4 SUMMARY



The origin of genus *Homo* marks the radical shift from earlier hominid species that displayed more primitive more ape- like characteristics than human- like characteristics. The genus, especially *Homo erectus* is also of importance since it is the first species to be able to migrate out of the African continent. Biological and physical adaptations by *Homo erectus* allowed the species to move and populate the rest of the world.

## 5.6 SELF- TEST QUESTIONS



1. How do we define the earliest members of the genus *Homo*?
2. When and where did *H. erectus* appear?
3. What sort of stone tools did *H. erectus* use?
4. How did a large brain aid *H. erectus* in its survival? What did the large brain allow them to do that earlier hominids could not do in expanding their territory beyond Africa?

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## 5.7 GLOSSARY



**Angular torus** a thickened bony ridge on the back of the parietal

**Occipital torus**, a ridge of bone that runs horizontally across the occipital

**Supraorbital torus** or the brow ridge on the front of the skull

## REFERENCES



Anton, S. C., & Swisher, C. C., III. (2004). Early dispersals of Homo from Africa. *Annual Review of Anthropology*, 33, 271–296.

Potts, R. (2004) Paleoenvironmental basis of cognitive evolution in great apes. *Am. J. Primatol.* 62,209–228.

Stanford Craig, Allen S. Allen & Anton, Susan C., (2006) *Biological Anthropology*. Upper Saddle River, NJ: Pearson Prentice Hall Publishers.

Vrba, E.S. 1985. Ecological and adaptive changes associated with early hominid evolution. In *Ancestors: the hard evidence* (ed. E. Delson), pp. 63-71. New York, Alan R. Liss.

Vrba, E.S., 1996. Climate, heterochrony, and human evolution. *Journal of Anthropological Research* 52 (1): 1-28.

Walker, A. and P. Shipman. 1996. *The wisdom of bones: in search of human origins*. New York, Vintage Books.

Ward, C. 2003. The evolution of human origins. *American Anthropologist*, 105, 77-88.

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## ORIGINS of MODERN HUMANS

### LECTURE SIX

#### 6.1 INTRODUCTION



Origin of modern humans is believed to have been a relatively recent evolutionary phenomenon. The current evidence which is both anatomical genetic and archaeological points to Africa as the origin of modern humans. The current best explanation for the beginning of modern humans is the Out of Africa Model that postulates a single, African origin for *Homo sapiens*. The major neurological and cultural innovations that characterized the appearance of fully modern humans has proven to be remarkably successful, culminating in our dominance of the planet at the expense of all earlier hominid populations.

There are two theories about the origin of modern humans: 1) they arose in one place — Africa and 2) pre-modern humans migrated from Africa to become modern humans in other parts of the world. Most evidence points to the first theory because:

- fossils of modern-like humans are found in Africa
- stone tools and other artifacts support African origin
- DNA studies suggest a founding population in Africa

#### 6.2 LECTURE OBJECTIVES



In this lecture students are expected to understand what modernity in humans‘ means. They are supposed to understand the theories that explain the origin of modern humans and the evidence available. Students are also expected to understand the process of migration and settlement from the continent of origin- Africa to the rest of the world.

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## 6.3 ORIGINS of MODERN HUMANS

### 6.3.1 Multiregional or Out of Africa?

One of the most hotly debated issues in the study of human origins focuses on the origins of modern humans, *Homo sapiens*. Roughly 100,000 years ago, the Old World was occupied by a morphologically diverse group of hominids. In Africa and the Middle East there was *Homo sapiens*; in Asia, *Homo erectus*; and in Europe, *Homo neanderthalensis*. However, by 30,000 years ago this taxonomic diversity vanished and humans everywhere had evolved into the anatomically and behaviorally modern form. The nature of this transformation is the focus of great deliberation between two schools of thought: one that stresses multiregional continuity and the other that suggests a single origin for modern humans.

#### Understanding the issue

The **Multiregional Continuity Model** contends that after *Homo erectus* left Africa and dispersed into other portions of the Old World, regional populations slowly evolved into modern humans. This model contains the following components:

- some level of gene flow between geographically separated populations prevented speciation, after the dispersal
- all living humans derive from the species *Homo erectus* that left Africa nearly two million-years-ago
- natural selection in regional populations, ever since their original dispersal, is responsible for the regional variants (sometimes called races) we see today
- the emergence of *Homo sapiens* was not restricted to any one area, but was a phenomenon that occurred throughout the entire geographic range where humans lived

The **Out of Africa Model** asserts that modern humans evolved relatively recently in Africa, migrated into Eurasia and replaced all populations which had descended from *Homo erectus*. Critical to this model are the following tenets:

- after *Homo erectus* migrated out of Africa the different populations became reproductively isolated, evolving independently, and in some cases like the Neanderthals, into separate species
- *Homo sapiens* arose in one place, probably Africa (geographically this includes the Middle East)
- *Homo sapiens* ultimately migrated out of Africa and replaced all other human populations, without interbreeding
- modern human variation is a relatively recent phenomenon

The multiregional view posits that genes from all human populations of the Old World flowed between different regions and by mixing together, contributed to what we see today as fully modern humans. The replacement hypothesis suggests that the genes in fully modern humans all came out of Africa. As these peoples migrated they replaced all other human populations with little or no interbreeding.

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To understand this controversy, the anatomical, archaeological, and genetic evidence needs to be evaluated.

## 6.3.2 ANATOMICAL EVIDENCE

Sometime prior to 1 million years ago early hominids, sometimes referred to as *Homo ergaster*, exited Africa and dispersed into other parts of the Old World. Living in disparate geographical areas their morphology became diversified through the processes of genetic drift and natural selection.

- In Asia these hominids evolved into Peking Man and Java Man, collectively referred to as *Homo erectus*.
- In Europe and western Asia they evolved into the Neanderthals.

Neanderthals lived in quasi isolation in Europe during a long, relatively cool period that even included glaciations. Neanderthals are distinguished by a unique set of anatomical features, including:

- a large, long, low cranial vault with a well-developed double-arched browridge
- a massive facial skeleton with a very projecting mid-face, backward sloping cheeks, and large nasal aperture, with large nasal sinuses
- an oddly shaped occipital region of the skull with a bulge or bun
- molars with enlarged pulp chambers, and large, often very heavily worn incisors
- a mandible lacking a chin and possessing a large gap behind the last molar
- a massive thorax, and relatively short forearms and lower legs
- although short in stature they possessed robustly built skeletons with thick walled limb bones
- long clavicles and very wide scapulas

By 130,000 years ago, following a prolonged period of independent evolution in Europe, Neanderthals were so anatomically distinct that they are best classified as a separate species — *Homo neanderthalensis*. This is a classic example of geographic isolation leading to a speciation event.

In contrast, at roughly the same time, in Africa, a body plan essentially like our own had appeared. While these early *Homo sapiens* were anatomically modern they were not behaviorally modern. It is significant that modern anatomy evolved prior to modern behavior. These early *sapiens* were characterized by:

- a cranial vault with a vertical forehead, rounded occipital and reduced brow ridge
- a reduced facial skeleton lacking a projecting mid-face
- a lower jaw sporting a chin
- a more modern, less robustly built skeleton

Hence, the anatomical and paleogeographic evidence suggests that Neanderthals and early modern humans had been isolated from one another and were evolving separately into two distinct species.

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## 6.3.3 ARCHAEOLOGICAL EVIDENCE

Even though Neanderthals and early *Homo sapiens* were distinguished from one another by a suite of obvious anatomical features, archaeologically they were very similar. Hominids of the Middle Stone Age of Africa (*H. sapiens*) and their contemporary Middle Paleolithic Neanderthals of Europe had artifact assemblages characterized as follows:

- little variation in stone tool types, with a preponderance of flake tools that are difficult to sort into discrete categories
- over long periods of time and wide geographical distances there was general similarity in tool kits
- a virtual lack of tools fashioned out of bone, antler or ivory
- burials lacked grave goods and signs of ritual or ceremony
- hunting was usually limited to less dangerous species and evidence for fishing is absent
- population densities were apparently low
- no evidence of living structures exist and fireplaces are rudimentary
- evidence for art or decoration is also lacking

The archaeological picture changed dramatically around 40-50,000 years ago with the appearance of behaviorally modern humans. This was an abrupt and dramatic change in subsistence patterns, tools and symbolic expression. The stunning change in cultural adaptation was not merely a quantitative one, but one that represented a significant departure from all earlier human behavior, reflecting a major qualitative transformation. It was literally a —creative explosion‡ which exhibited the —technological ingenuity, social formations, and ideological complexity of historic hunter-gatherers.‡ (Klein 2000)

The appearance of fully modern behavior occurred in Africa earlier than anywhere else in the Old World, but spread very quickly, due to population movements into other geographical regions. The Upper Paleolithic lifestyle, as it was called, was based essentially on hunting and gathering. So successful was this cultural adaptation that until roughly 11,000 years ago, hominids worldwide were subsisting essentially as hunter-gatherers.

In the Upper Paleolithic of Eurasia, or the Late Stone Age as it is called in Africa, the archaeological signature stands in strong contrast to that of the Middle Paleolithic/Middle Stone Age. It was characterized by significant innovation:

- a remarkable diversity in stone tool types
- tool types showed significant change over time and space
- artifacts were regularly fashioned out of bone, antler and ivory, in addition to stone
- stone artifacts were made primarily on blades and were easily classified into discrete categories, presumably reflecting specialized use
- burials were accompanied by ritual or ceremony and contained a rich diversity of grave goods
- living structures and well-designed fireplaces were constructed
- hunting of dangerous animal species and fishing occurred regularly higher population densities
- abundant and elaborate art as well as items of personal adornment were widespread

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- raw materials such as flint and shells were traded over some distances

*Homo sapiens* of the Upper Paleolithic/Late Stone Age were quintessentially modern in appearance and behavior. Precisely how this transformation occurred is not well understood, but it apparently was restricted to *Homo sapiens* and did not occur in Neanderthals. Some archaeologists invoke a behavioral explanation for the change. For example, Soffer (1990) suggests that changes in social relations, such as development of the nuclear family, played a key role in bringing about the transformation.

Klein (2000) on the other hand, argues that it was probably a biological change brought about by mutations that played the key role in the emergence of behaviorally modern humans. His biologically based explanation implies that a major neural reorganization of the brain resulted in a significant enhancement in the manner in which the brain processed information. This is a difficult hypothesis to test since brains do not fossilize. But it is significant that no changes are seen in the shape of the skulls between earlier and later *Homo sapiens*. It can only be surmised from the archaeological record, which contains abundant evidence for ritual and art, that these Upper Paleolithic/Late Stone Age peoples possessed language abilities equivalent to our own. For many anthropologists this represents the final evolutionary leap to full modernity.

Shortly after fully modern humans entered Europe, roughly 40,000 years ago, the Neanderthals began a fairly rapid decline, culminating in their disappearance roughly 30,000 years ago. Neanderthals were apparently no match for the technologically advanced fully modern humans who invaded Europe and evidence for interbreeding of these two types of hominids is equivocal.

## 6.3.4 GENETIC EVIDENCE

Investigation of the patterns of genetic variation in modern human populations supports the view that the origin of *Homo sapiens* is the result of a recent event that is consistent with the Out of Africa Model. Studies of contemporary DNA, especially mitochondrial DNA (mtDNA) which occurs only in the cellular organelles called mitochondria, reveal that humans are astonishingly homogeneous, with relatively little genetic variation (Cann *et al.* 1987).

The high degree of similarity between human populations stands in strong contrast to the condition seen in our closest living relatives, the chimpanzees (Cavalli-Sforza, 2000). In fact, there is significantly more genetic variation between two individual chimpanzees drawn from the same population than there is between two humans drawn randomly from a single population. Furthermore, genetic variation between populations of chimpanzees is enormously greater than differences between European, Asian and African human populations.

In support of an African origin for *Homo sapiens* the work of Cann has demonstrated that the highest level of genetic variation in mtDNA occurs in African populations. This implies that *Homo sapiens* arose first in Africa and has therefore had a longer period of time to accumulate genetic diversity. Using the genetic distance between African populations and others as a measure of time, they furthermore suggested that *Homo sapiens* arose between 100,000 and 400,000 years ago in Africa.

The low amount of genetic variation in modern human populations suggests that our origins may reflect a relatively small founding population for *Homo sapiens*. Analysis of mtDNA by Rogers and

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Harpending (1992) supports the view that a small population of *Homo sapiens*, numbering perhaps only 10,000 to 50,000 people, left Africa somewhere between 50,000 and 100,000 years ago.

Scientists recently succeeded in extracting DNA from several Neanderthal skeletons (Krings *et al* 1999). After careful analysis of particularly the mtDNA, but also some nuclear DNA, it is apparent that Neanderthal DNA is very distinct from our own. In assessing the degree of difference between DNA in Neanderthals and modern humans, the authors suggest that these two lineages have been separated for more than 400,000 years.

Although in its infancy, such genetic studies support the view that Neanderthals did not interbreed with *Homo sapiens* who migrated into Europe. It is, therefore, highly likely that modern humans do not carry Neanderthal genes in their DNA.

## 6.3.5 Additional Considerations

- The chronology in the Middle East does not support the Multiregional Model where Neanderthals and anatomically modern humans overlapped for a long period of time.
- Cave sites in Israel, most notably Qafzeh and Skhul date to nearly 100,000 years and contain skeletons of anatomically modern humans. Furthermore, Neanderthal remains are known from sites such as the 110,000-year-old Tabun cave, which predates the earliest *Homo sapiens* by about 10,000 years in the region.
- The presence of Neanderthals at two other caves in Israel, Amud and Kebara, dated to roughly 55,000 years means that Neanderthals and *Homo sapiens* overlapped in this region for at least 55,000 years. Therefore, if *Homo sapiens* were in this region for some 55,000 years prior to the disappearance of the Neanderthals, there is no reason to assume that Neanderthals evolved into modern humans.
- Archaeological evidence from Europe suggests that Neanderthals may have survived in the Iberian Peninsula until perhaps as recently as 30,000 to 35,000 years ago. Fully modern humans first appear in Europe at around 35,000-40,000 years ago, bringing with them an Upper Paleolithic tool tradition referred to as the Aurignacian. Hence, Neanderthals and fully modern humans may have overlapped for as much as 10,000 years in Europe. Again, with fully modern humans on the scene, it is not necessary to have Neanderthals evolve into modern humans, further bolstering the view that humans replaced Neanderthals.
- The situation in southern France is, however, not quite as clear. Here, at several sites, dating to roughly 40,000 years there is evidence of an archaeological industry called the Châtelperronian that contains elements of Middle and Upper Paleolithic artifacts. Hominids from these sites are clearly Neanderthals, sparking speculation that the Châtelperronian is an example of Neanderthals mimicking the culture of modern humans. The lack of anatomical intermediates at these sites, suggests that if Neanderthals did encounter and borrow some technology from *Homo sapiens*, they did not interbreed.
- A potential 24,500-year-old Neanderthal/*sapiens* hybrid was announced from the site of Lagar Velho, Portugal. This 4-year-old has a short, squat body like a Neanderthal, but possesses an anatomically modern skull. There are a number of problems with interpreting this find as a Neanderthal/*sapiens* hybrid. First of all, as a hybrid it should have a mixture of traits throughout its body and not possess the body of a Neanderthal and skull of a modern human. For example, if we look at hybrids of lions and tigers they do not possess the head

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of one species and the body of the other, but exhibit a morphological mixture of the two

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species. Secondly, and more importantly, acceptance of this specimen as a hybrid would suggest that Neanderthal traits had been retained for some 6,000 to 10,000 years after Neanderthals went extinct, which is highly unlikely. This is theoretically unlikely since Neanderthal traits would have been genetically swamped by the *Homo sapiens* genes over such a protracted period of time.

- Proponents of the Multiregional Model, such as Milford Wolpoff, cite evidence in Asia of regional continuity. They see an evolutionary link between ancient *Homo erectus* in Java right through to Australian aborigines. A possible problem with this view is that recent dating of late surviving *Homo erectus* in Indonesia suggests that they survived here until 50,000 years ago, which is potentially when fully modern humans may have arrived in the region from Africa.
- China may contain the best evidence for supporting the Multiregional Model. Here there are discoveries of a couple of skulls dated to roughly 100,000 years ago that seem to possess a mixture of classic *Homo erectus* and *Homo sapiens* traits. Better geological dating and more complete specimens are needed to more fully assess this possibility.

## 6.4 SUMMARY



For the moment, the majority of anatomical, archaeological and genetic evidence gives credence to the view that fully modern humans are a relatively recent evolutionary phenomenon. The current best explanation for the beginning of modern humans is the Out of Africa Model that postulates a single, African origin for *Homo sapiens*. The major neurological and cultural innovations that characterized the appearance of fully modern humans has proven to be remarkably successful, culminating in our dominance of the planet at the expense of all earlier hominid populations.

## 6.5 SELF- TEST QUESTIONS



1. How does the Multiregional Continuity Model explain the origin of modern humans?
2. The Out of Africa model posits that there was a recent African origin for all humanity, explain.
3. How is anatomical evidence used to explain the origin of modernity in humans?

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4. What were the technological differences/ similarities between MSA *H. sapiens* in Africa and Middle Paleolithic Neanderthals?
5. Neanderthals and modern humans in Europe may have interbred, explain.

## REFERENCES



- Cann, R.L., M. Stoneking, and A.C. Wilson. 1987. —Mitochondrial DNA and human evolution.‖ *Nature*, 325:32-36. Cavalli-Sforza, L.L. 2000. *Genes, Peoples, and Languages*. New York. North Point Press.
- Klein, R.. 1999. *The Human Career*. Chicago. University of Chicago Press.
- Klein, R. 2000. —Archaeology and the evolution of human behavior.‖ *Evolutionary Anthropology*, 9:17-36.
- Krings, M; Stone, A; Schmitz, R W; Krainitzki, H; Stoneking, M; Paabo, S. (1997). Neandertal DNA sequences and the origin of modern humans. *Cell* 90 19-30.
- Rogers, A. R. and Harpending, H. C. 1992. Population growth makes waves in the distribution of pairwise genetic differences. *Molecular Biology and Evolution*, 9:552{569.
- Soffer, O. 1990. —Before Beringia: Late Pleistocene biosocial transformations and the colonization of northern Eurasia.‖ In: —Chronostratigraphy of the Paleolithic in North, Central, East Asia and America.‖ *Novosibirisk. Acad. Of Sci. of the USSR*.
- Stringer , C. and R. McKie. 1996. *African Exodus: The Origins of Modern Humanity*. New York: Henry Holt.
- Tattersall, I. and J.H. Schwartz. 1999. —Hominids and hybrids: The place of Neanderthals in human evolution.‖ *Proc. Natl. Acad. Sci. USA*, 96:7117-7119.
- Wolpoff, M.H. and R. Caspari. 1996. *Race and Human Evolution: A Fatal Attraction*. New York. Simon and Schuster.
- Wolpoff, M.H., J. Hawks, D.W. Frayer and K. Hunley. 2001. —Modern human ancestry at the peripheries: A test of the replacement theory.‖ *Science*, 291:293-297.

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## BEHAVIOR and EVOLUTION of EARLY HOMININS

### LECTURE SEVEN

#### 7.1 INTRODUCTION



Early hominines display an ability to create and use tools. However, it is worth pointing out that humans are not the only animals that have the ability to fashion and use tools, since chimpanzees have been documented using tools. Humans however no matter the form of economic production that they undertake are the only animals that have become dependent on the fruits of technology. For this reason, and because archaeological record provides good evidence for the evolution of technology, it has become important to consider how stone tools relates to early hominine evolution and at what point did they become technology dependent.

From the archaeological and contemporary evidence, many forms of materials can be used as tools, such as stone, wood, bone and antler, bark and leaves. However according to preservation issues, only implements made from stone are likely to be recovered in the archaeological record. Apart from the preservation bias, stone tools can perform multiple activities such as slicing, scrapping, hammering and in making other tools such as digging sticks and spears. Thus, a record of stone-tool technologies in the earliest human groups therefore provides an important, although not complete insight to subsistence activities.

#### 7.2 LECTURE OBJECTIVES



In this lecture students are expected to understand the cultural and technological changes that have made humans successful in conquering the different environmental settings. Students are expected to understand various technological changes over time and how they have been a product of changes in brain sizes and modernity. Students are also expected to understand the regional variations that exist as far as technology is concerned

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## 7.3 THE ARCHAEOLOGY OF STONE TOOLS

Stone tools have been collected for many years by both amateurs and professionals in various parts of the world. The focus of the collection was to classify the stone tools as the phenomena in themselves with great emphasis on classification types. This focus has however changed in the contemporary studies with the strong interest being on studying artifacts within the subsistence context of early hominines. In extension, the focus seeks to answer questions such as: how broad was the diet? Specifically to what extent was an important subsistence activity? Were the subsistence activities including a —home base? How did the hominines exploit their range, and how large it was (Lewin and Foley 2004).

According to Grahame Clark, stone tool assemblages have been classified into five categories, modes that are defined by characteristic artifacts in them. These categories appear sequentially through time, but may overlap when earlier modes persist after the appearance of later modes. Mode 1 technology is based on simple chopping tools that are made by knocking a few flakes off a cobble. Mode 2 is characterized by tools that require more extensive conceptualization and preparation and in particular the bifacially flaked handaxes. Mode 3 is characterized by large cores that are pre-shaped by the removal of large flakes and then used as a source of more standardized flakes that are retouched to produce a large range of artifacts. Mode 4 technology is characterized by narrow stone blades struck from a prepared core. Mode 5 consists of microliths technology, which constitutes the production of small, delicate artifacts.

Thus Clarke's classification system enables the description of the characteristics of archaeological assemblages and not of the archaeological time periods. For example Mode 1 technology appears in Africa around 2.6 million years ago and persisted until historical times. Mode 4 blades on the other hand were produced in Africa for over 100,000 years, but do not appear in the European archaeological record until 40,000 years ago. This deviation can probable be explained by the dynamics of the adaptation and migration patterns among the hominines.

The time periods and Modes mentioned above are explained using different terminologies in sub-Saharan Africa and Asia. This can be attributed to the development of archaeology in the continents as compared to Europe. Lewin and Foley (2004) point out that the time before the appearance of agriculture and iron is known as the Stone Age. The Stone Age is divided into: the Early Stone Age (ESA), the Middle Stone Age (MSA), and the Late Stone Age (LSA). In North Africa and Europe the stone tools periods before the Neolithic are referred to as the Paleolithic and are divided into three stages are somewhat equivalent to the African Stone Age; the lower Paleolithic, the Middle Paleolithic, and the Upper Paleolithic. These periods have been defined mainly according to cultural evolution though time.

With the elasticity of stage boundaries, technology development in mind, the stone tool stages unfolded as follows. The beginning of the ESA corresponds with the first appearance of Mode 1 tools, 2.6 million years ago; the ESA includes the first appearance of Mode 2, approximately 1.5

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million years ago, and terminates with the first appearance of prepared cores (Mode 3), which also

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marks the beginning of the MSA, 300,000 years ago. In most cases Mode 4 is mostly associated with the Upper Paleolithic and the LSA, although some elements can be found occasionally in the MSA.

## 7.4 THE EARLIEST KNOWN TOOLS

### 7.4.1 THE OLDOWAN

The oldest stone tools in the archaeological record are dated to approximately 2.6 million years ago, from the sites in the Lower Omo Valley, Hadar and Gona region in Ethiopia, and in the western shore of Lake Turkana. The artifacts from the Lower Omo Valley are atypical in that they are small quartz pebbles shattered to produce sharp-edged implements. This is unlike most tools dating from the period 2.6 million to 1.5 million years ago that are mainly made from lava cobbles, which consist what is mainly referred to as the core tools and small sharp flakes. In general, the technology is known as **Oldowan** after Olduvai Gorge, Tanzania (**Fig 6.1**). The technology which is Mode 1 was defined on the basis of the artifact assemblages found in bed I and lower bed II at Olduvai dating at around 1.9 to 1.6 million years old.

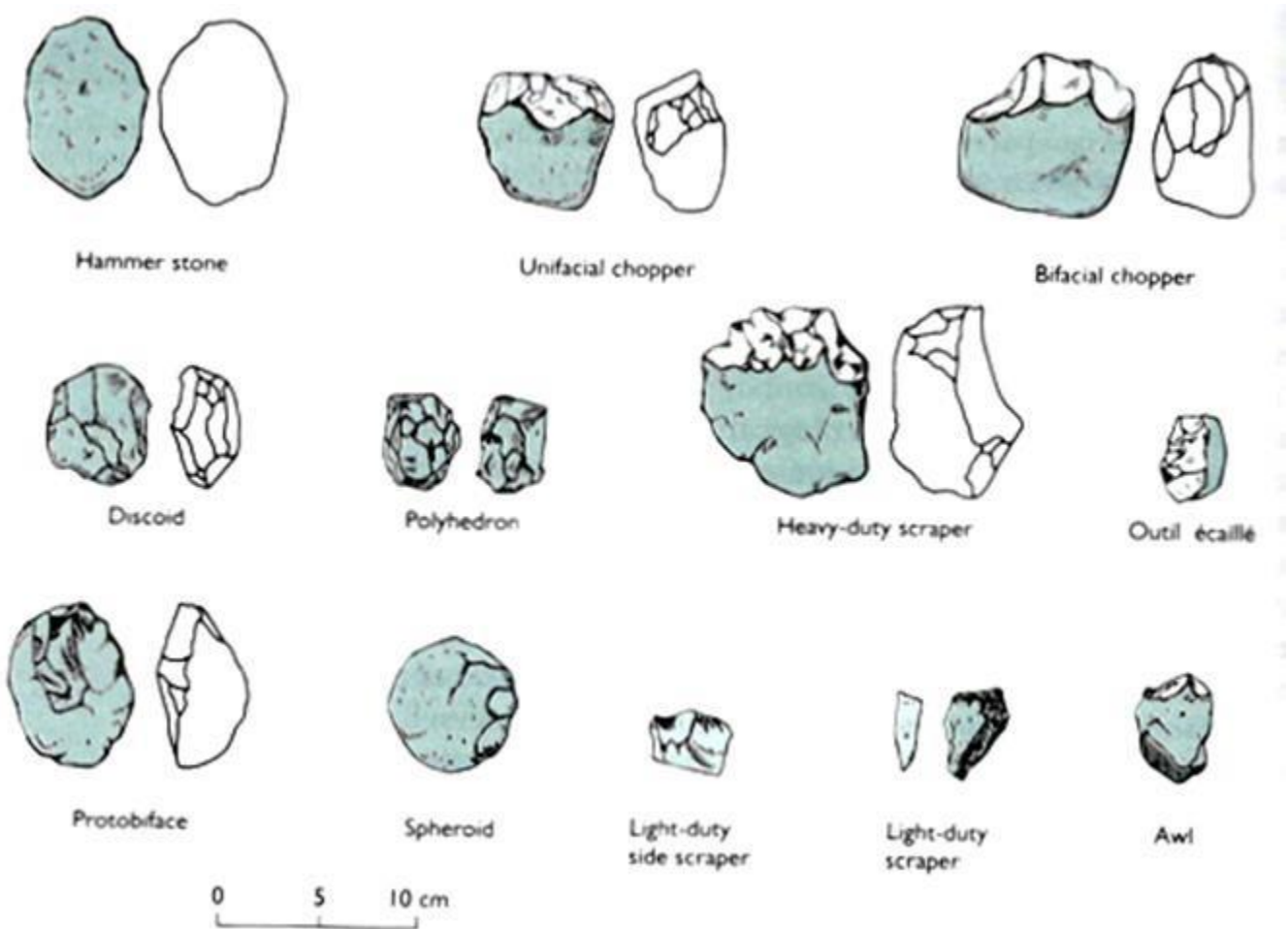
The artifacts fall into four categories.

- Tools, which include types such as scrapers, choppers, discoids, and polyhedrons
- Utilized pieces, such as large flakes produced in the manufacture of tools, having sharp edges useful for cutting
- Waste, which is small pieces produced in the manufacture or retouching of tools and utilized piece
- Manuports, which are pieces of rock carried to a site but modified.

It is important to note that the typical Oldowan seem to have been produced opportunistically unlike the later technologies that show clear mental mapping and planning in the production. The different forms tended to flow into one another typologically, and seem to have been produced for a given purpose at the specific time. The tool production process in the Oldowan was the flakes. The makers were deliberately producing flakes, and the cores were by-products of this process. The variation in the core and flake sizes can be attributed to the variation in size of the available raw materials.

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**Fig 6.1.** Oldowan artifacts: the manufacture of these pebble tools requires considerable skills (courtesy of Lewin & Foley 2006, 314)

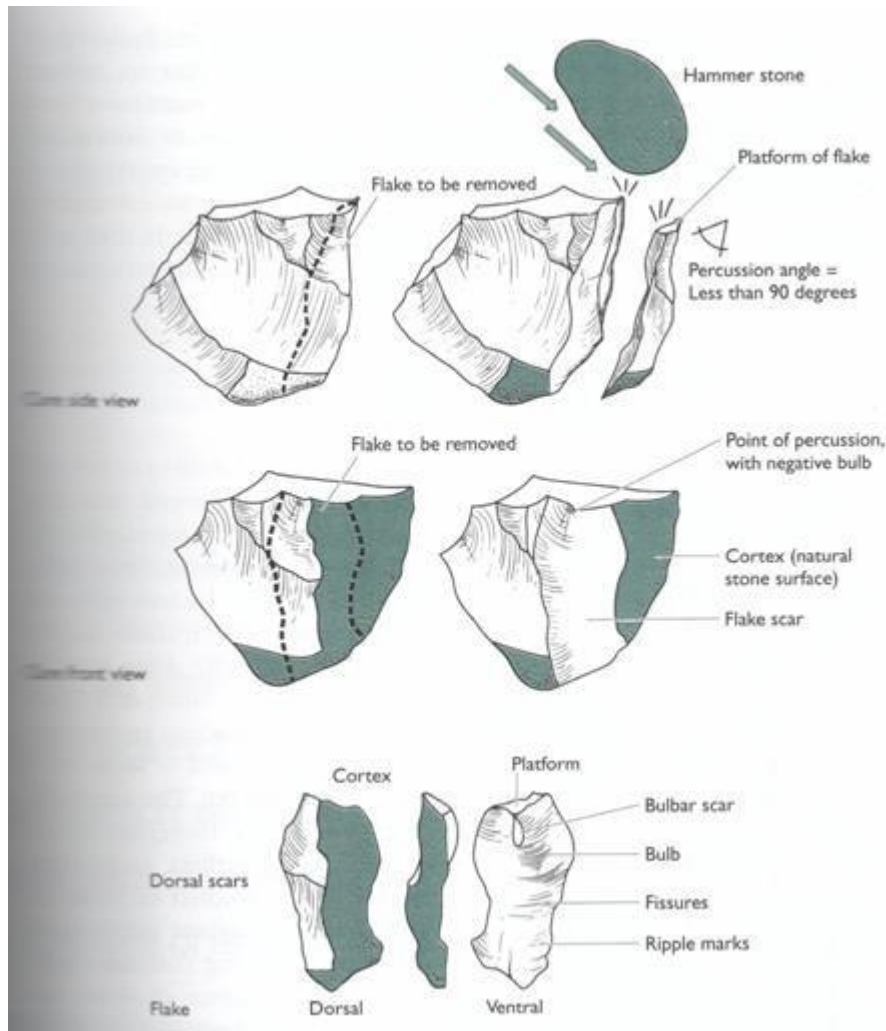
Flakes could be used in de-fleshing, shearing, cutting through meat and modifying other tools such as digging sticks. Hammerstones and heavy cores could be used to open up bones for marrow or brain. Other forms of tools such as antlers, broken bones and digging sticks could have been used in the Oldowan but due to the vagaries of preservation it is hard to clearly locate them in the archaeological record.

## SKILLFUL OLDOWAN TOOL MAKERS

The hominine ability to produce tools represents a technological revolution of some sorts. Although the Oldowan is rather crude, the production of flakes is a complicated process that represented a shift and improvement in their brain wiring. For it to be successful three conditions must be met by the stone knapper producing flakes (**Fig 6.2**). First, the core must have an acute edge, one less than 90 degrees, near which the hammer can strike. Secondly, the core must be struck with a glancing blow about 1 cm from the acute edge. Third, the blow must be directed through an area of high mass, such as a ridge or a bulge.

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**Fig. 6.2.** Diagnostic features of flaking by percussion: providing sharp usable flakes requires the delivery of forceful blows at the correct angle and at the correct location on the core. The flakes produced in this manner have certain features produced by conchoidal fracture (courtesy of Lewin & Foley 2006, 317)

About 1.6 million years ago, a new form of Oldowan industry emerged; this technology was named the Developed Oldowan by Mary Leakey. It is characterized by a smaller percentage of choppers, and a greater abundance of spheroids, subspheroids, and small scrapers. For the first time, bifacial tools appear in this industry, including prototype handaxes and cleavers, which later came to characterize the Acheulian industry. This industry or others similar to it persisted in Africa until around half a million years ago and in eastern Asia until as late as 200,000 years ago.

## Who Made the Tools?

Within the period 2.6 million to 1.5 million years ago, several hominine species (*Homo* and *Australopithecus*) lived as contemporaries. From the available evidence it looks like *Homo* was the tool makers although it does not rule out the other species. The earliest evidence of stone-tool making coincides with the first appearance of *Homo* approximately 2.5 million years ago.

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## New Technologies

The evolution of *Homo* brought many changes in the biology of our direct ancestors. Variations in life- history factors, in social structure, and in subsistence patterns combined to make species —more humanl that earlier species of *Homo* or *Australopithecus*. Of great importance was the development of meat as a significant component of the diet especially in providing the stability and richness of energy resources and in allowing new habitats to be exploited (Lewin and Foley 2004). *Homo erectus* was the first hominin to move beyond the African continent. These developments were accompanied by significant enhancement of stone- tool technologies.

### 7.4.2 THE ACHEULIAN TECHNOLOGY

When compared to the Oldowan, the subsequent technology in the archaeological record shows significant innovations in the appearance of the Acheulian assemblage, which id Mode 2 technology. According to Asfaw *et al.* (1992), the earliest known collection of this assemblage comes from Konso- Gardula, Ethiopia, and is 1.4 million years ago. However, the name Acheulian derives from the site of St. Acheul, in northern France where many examples of handaxes were collected in the last century. The innovation consisted of the introduction of larger tools- known as handaxes, picks, and cleavers- than seen in the Oldowan. Compared to the Oldowan choppers, Acheulian handaxes required a higher level of cognitive ability in the conceptualization of the end product and its manufacture.

The tools are generally associated with *Homo erectus*. Although the earliest known *Homo erectus* fossils appear in the archeological record close to two million years ago, the earliest known Acheulian technology appears around half a million years later. The gap between the appearance of the species and the technology can be explained by one, that the innovation was cultural, with later *Homo erectus* populations inventing the new tool technology after having employed the simpler technology Oldowan technology for half a million years. Secondly, the Acheulian technology may have been a *Homo erectus* innovation.

Once the large, bifacial handaxe appeared, it remained a characteristic of Acheulian assemblages for a very long time, in both Africa and Eurasia. Production became refined through the millennia, so that some late examples appear finely made, compared to the crude earlier specimens. Defining character of the Acheulian is that it included an increased reliance on more detailed preparation of the piece upon which the handaxe was then made. This core preparation, one method of which was is known as the Levallois technique became especially dominant in the Middle Stone Age and the Middle Paleolithic technologies (Mode 3 technologies). Thus, the Acheulian was marked by a long continuous technological continuity maintained through a very long period of time.

Acheulian assemblages are known form many sites in Africa. Such sites includes at Olorgesailie (700,000), Muguruk, Kapthurin in Baringo, Chesowanja (McBrearty 1999). The industry persisted until at around 200,000 years ago, when it is superseded by Middle Stone Age (Middle Paleolithic) assemblages. The earliest Acheulian site outside of Africa is dated to approximately 1 million years

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ago at Ubeidiya, west of the Sea of Galilee, in Israel. The site seems to be on the possible migration route out of Africa into Asia. Acheulian in Europe is present at the sites of Isernia in Italy at 700,000 and Vertesszöllos in Hungary and Arago in France both dating over 300,000 years ago. Thus the earliest Acheulian assemblages are located in Africa but later sites are found in western Asia, southern Asia, and Europe.

## **The Function of the Acheulian**

The function of the Acheulian has been in the debates for long. The main hypothesis and the one that seems to hold more weight is that they used as axes or heavy-duty knives. The tools may also have been used to slice tough hide from animals. The combination of weight and relatively sharp edges gives them greater efficacy than the small, sharp flakes. Microwear studies have shown that handaxes were used for many functions ranging from meat and bone to wood and hide.

The end of the Acheulian industries which occurred from 300,000 to around 200,000 years ago throughout the Old World marked the end of these tool assemblies that had few artifact types and enjoyed enormous longevity. The end of the Acheulian brought the Early Stone Age (lower Paleolithic) to a close and marked the beginning of the industries of the Middle Stone Age (Middle Paleolithic). The MSA lasted from around 300,000 to 40,000 years ago.

## **7.4.3 THE MIDDLE STONE AGE**

The Middle Stone Age in Africa was between Early Stone Age and Late Stone Age. It began around 300,000 years ago and ended around 50,000 years ago (McBrearty & Brooks, 2000). The technology was in Africa and the Near East, while in Europe, albeit much later the corresponding technology is referred to as Middle Paleolithic. The technology is characterized by the production of flakes and flake-blades, some of them retouched to form scrapers, knives, points or backed pieces. It is associated with anatomically modern or almost modern *Homo sapiens*. Early physical evidence comes from Omo and Herto, both in Ethiopia and dated respectively at c. 195 ka and at c. 160 ka.

The stage is based on the characteristics of tool assemblages in the different geographical regions. This means that their chronological equivalence is not exact and neither is there an exact equivalence of artifact types. The archaeology associated with the origin of modern humans relates to the development of Modes 3 and 4 technologies. These terms refer to the technologies specifically, but there is also the important aspect of how the technology is related to other aspects of behavior such as hunting strategies or spatial patterns.

According to Lewin and Foley the end of Modes 1 and 2 at 250,000 years ago saw the end of innovation-poor, long lasting stone tool industries. With the beginning of the MSA, the number of identifiable tool types quadrupled, reaching about 40. With the Later Stone Age (Mode 4), beginning 40,000 years ago, the number of tools more than doubled again, to as many as 100. In Modes 3 and 4, regional variations became quite distinctive with marked variability through space and time. In addition to the variety in tools, raw materials such as bone, ivory and antler that were not used as

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much in the Oldowan and Acheulian became more important especially in the Later Stone Age.

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The Middle Stone Age (Mode 3) technologies were characterized by the predominance of the prepared core technique, such as the Levallois technique which appeared earlier. The Levallois technique involved preparation of a large core so that it has a flat upper surface and convex lower surface. The considerable force required to detach the broad flakes is applied by bringing the striking platform of the core down sharply at an angle on an anvil. The relatively large, thin flakes conform to the shape of the outline of the core. Many flakes of similar form may be produced by repeatedly striking around the edge of the core until it is virtually all used up. The Levallois technique is much more economical of all the raw material than earlier flaking techniques producing many more centimeters of working edge for each kilogram of core (Lewin and Foley 2004).

Once produced the, the flakes may then be fashioned further to give what some archaeologists identify as approximately 40 different implements, each with its own putative cutting, scrapping, or piercing function. Some of the identified types may represent stages in the manufacture of other artifact types or the products of repeated resharpening, meaning that they may not be real tool types. Some variations exist in the MSA which has led to the development of various local names. In Africa and Asia there are many regional local styles such as Still Bay, Howiesons Poort, etc (**Fig. 6.3**).

## **Early Development**

During the Acheulian to MSA transition the Middle Awash valley of Ethiopia and the Olorgesailie basins of Kenya constituted a major center for behavioral innovation. It is likely that the large terrestrial mammal biomass of these regions supported substantial human populations with subsistence and manufacturing patterns similar to those of ethnographically known forager. According to McBrearty and Brooks (2000), MSA innovations did not appear suddenly in a single locality but different sites at different times. Blades and backed pieces from the Kapthurin, Twin Rivers and Kalambo Falls sites in Zambia dating anywhere between 300 Kya to 75 Kya indicate a suite of new behaviors. Blade technology of the production of points is also evident in the MSA are evident in Africa some dating to 235,000 years ago.

The later MSA is also characterized by the presence of microliths. This is otherwise surprising since it was earlier though that microliths were a characteristic of the Later Stone Age and the Upper Paleolithic Europe. According to McBrearty and Brooks (2000), the site of Mumba in Tanzania has evidence of microliths production 65,000 years ago. This is also the case in at Border Cave's Howieson's Poort industry.

At 70 - 55 kya and Barham believes that syntactic language was one behavioral aspect that allowed these MSA people to settle in the tropical forests of the Congo. A high level of technical competence is also indicated for the c. 280 ka blades recovered from the Kapthurin Formation, Baringo. McCall (2006) contend that distinct technological changes in lithic style between the MSA I period (c. 110 – 115 ka) and the MSA II (c. 94 -85 ka) at Klasies in the Western Cape is associated with cognitively modern behaviour.

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During the MSA, there is evidence for increased diversity of food exploited. A number of MSA coastal sites show exploitation of marine mammals, fish, shellfish, and tortoises earlier than 40,000 years ago (Stanford *et al.* 2006)

Although some regional industries have been identified, detailed classification of the Middle Stone Age assemblages is generally avoided: this is mainly because there are few long sequence sites and chronological controls, especially for open air location, are poor.

The transition dates for the Middle and Later Stone Ages is debatable, especially when dealing with the regional variance in dates. In the Southern Africa, it has been dated to around 21,000 BP. These dates are based on data from sites such as Boomplaas and Rose Cottage Cave.

## 7.4.4 LATER STONE AGE

The Later Stone Age are distinguished from the MSA by a greater production of blades and microliths. A number of blades could be taken off a prepared core in a systematic manner. Microliths are small, flaked tools that appeared after 25,000 years ago in most regions. From the way they were made, it appears they were designed to be attached to wood or bone. Arrow heads are an example of the microliths and appear at around 13,000 -10,000 years ago (Stanford *et al.*).

The Later Stone Age is also characterized by the vastly greater use of tools made from bone, ivory, antler and shell. The tools were ground, polished and drilled to form objects such as harpoons and awls.

Also in the Later Stone Age, there is substantial evidence on emphasis on the use of ostrich eggshell and bone beads, bored stone digging stick weights, tortoiseshell bowls, ostrich eggshell containers and bone points.

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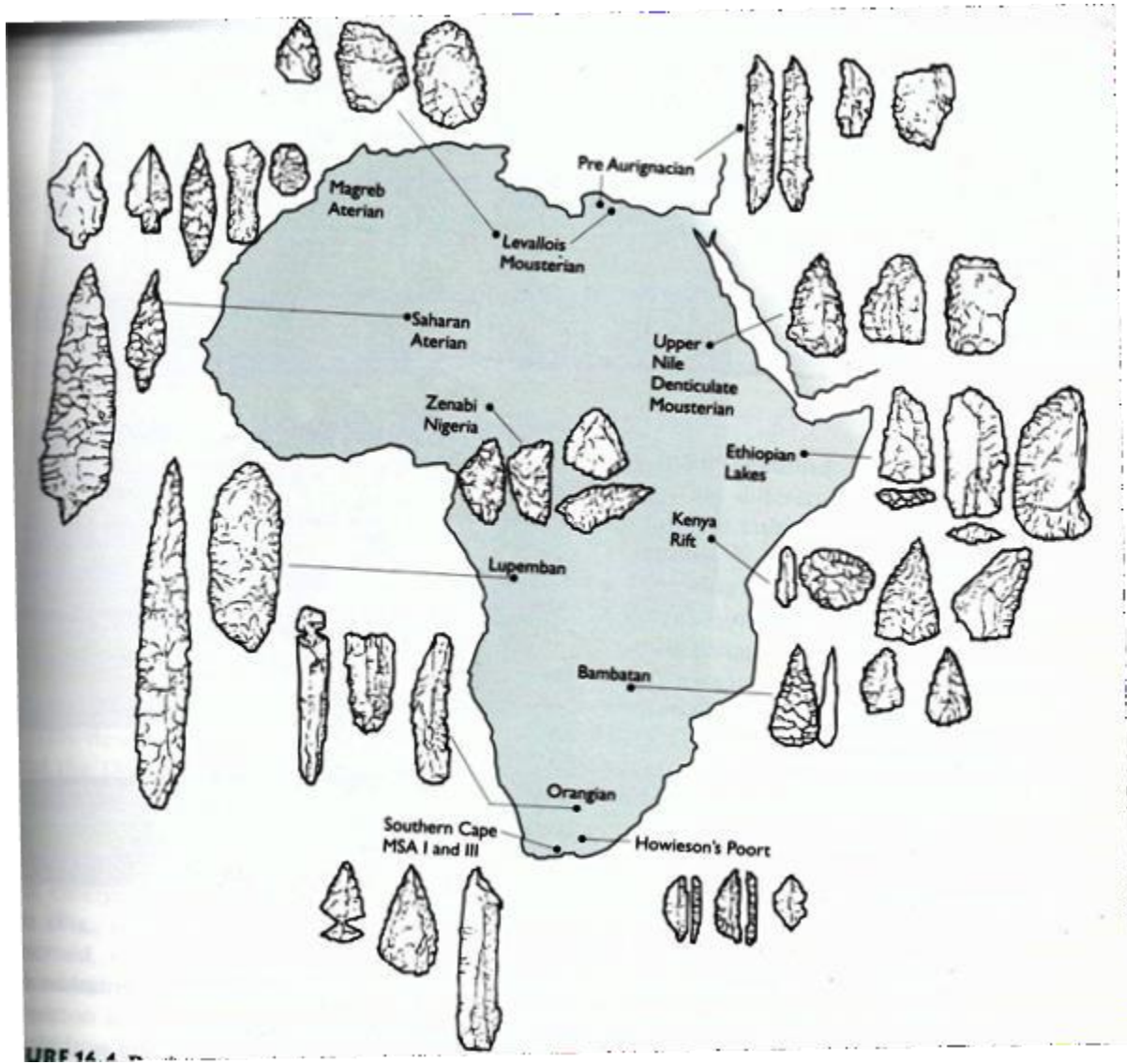


Fig. 6.3. Regional Variation in Middle Stone Age of Africa (courtesy J.D. Clark)

## 7.5 SUMMARY



The invention, use and refinement of stone tools by various hominids allowed them to conquer the

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environment, move around the world and improve their lot. Technology in tool production went

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through processes of refinement over time, although there is evident for some retained technology. The retention of some of the technology such as Acheulian can be attributed to suitability and relevance of that technology to the individuals that used it.

## 7.6 SELF- TEST QUESTIONS



1. How does the MSA differ with the ESA?
2. Why is it difficult to determine who made particular tools over given period of time?
3. What are the defining characteristic differences between the MSA and the LSA?
4. How are various technological changes in the stone tool technology a sign for changes in the cognition abilities?

## REFERENCES



Christopher Stuart Henshilwood and Benoît Dubreuil (2001) The Still Bay and Howiesons Poort, 77– 59 ka Symbolic Material Culture and the Evolution of the Mind during the African Middle Stone Age. *Current Anthropology* Vol. 52 No. 3.

Lewin, Roger & Foley Robert, A. (2004). *The Principles of Human Evolution*. Malden MA: Blackwell Science Publishers.

McBrearty Sally. (1999) –the Archaeology of Kapthurin Formation in Andrews P & Banham P( eds.). *Late Cenozoic Environments and Hominid Evolution: a Tribute to Bill Bishop*. London: Geological Society, 143- 156.

McBrearty Sally & Brooks Allison (2000) The Revolution That Wasn't: A New Interpretation Of The Origin Of Modern Human Behavior. *Journal of Human Evolution* (2000) 39, 453–563

Stanford Craig, Allen S. Allen & Anton, Susan C. *Biological Anthropology*. Upper Saddle River, NJ: Pearson Prentice Hall Publishers.

Grant S. McCall (2006) Multivariate perspectives on change and continuity in the Middle Stone Age lithics from Klasies River Mouth, South Africa. *Journal of Human Evolution*, 51 (4) 429-439.

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## AFRICAN PRE- HISTORIC ART

### LECTURE EIGHT

#### 8.1 INTRODUCTION



The account of the origins of art is a very long one marked less by change than consistency. Art, as the product of human creativity and imagination, includes poetry, music, dance, and the material arts such as painting, sculpture, drawing, pottery, and bodily adornment. The first human artistic representations, markings with ground red ocher, seem to have occurred about 100,000 B.C. in African rock art. African art encompasses a diverse range of ceramic and cement sculptures, sculpted and carved animal bones, urban murals and rock images. The art forms depict African perspectives of the world and their ideologies.

#### 8.2 LECTURE OBJECTIVES



In this lesson students will be introduced to prehistoric African art. This includes material art and also rock paintings body adornments etc. students are expected to understand the ages associated with the art, their distribution, threats associated with the art and the comparison with art from other continents.

#### 8.3 AFRICAN ART

African prehistoric art refers to Africa's Stone Age art. Africa has the greatest number of specimens of prehistoric art amongst all continents. Art does not serve as just a decoration item but is integral with the lifestyle of people. Art, as the product of human creativity and imagination, includes poetry, music, dance, and the material arts such as painting, sculpture, drawing, pottery, and bodily adornment.

It is during the Upper Paleolithic that a wider Range of cultural artifacts including complex cave art began to be produced. In the sphere of human evolution and modernity, the period is characterized by exponential. The earliest form of African art consisted of pictographs and petroglyphs which are

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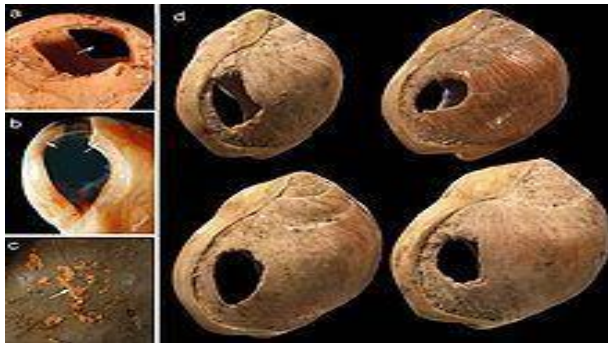
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more than 20, 000 years old. Rock art forms have been found in South Africa and Namibia. The rock surfaces have depicted animal and human activities.

By 20,000 B.C., humans had settled on every continent except Antarctica. The earliest human occupation occurs in Africa, and it is there that we assume art to have originated. African rock art from contain examples of geometric and animal representations engraved and painted on stone. Accepted dates for the earliest African rock art are a matter of debate. Current dates suggest art form to have been present between 30,000 and as early as 200,000 years ago. According to Chris Heshinwood (2006), archaeological evidence shows that African rock painting is as old as 70,000 years ago. His evidence is based on archaeological material recovered at Blombos cave in South Africa which comprised a collection of artifacts. In particular, there were two pieces of ochre rock incised with abstract geometric patterns (**Fig. 1**), and a series of beads made from *Nassarius kraussianus* shell (**Fig 2**). Other art forms of significant importance from Blombos of included shell beads dating from 70,000 – 75,000 BCE.



**Fig. 1.** One of the engraved stones at Blombos dating to 70,000 BCE.



**FIG. 2:** *Nassarius kraussianus* shell beads

In rock painting, colouring materials used for the rock paintings were obtained from minerals in the soil, ground into powder and mixed most probably with animal fats. Minerals containing iron, such as ochre, give many colours. On the other hand black paint was made from manganese and also from charcoal. There is also a possibility that colours such as blue were made from vegetable dyes.

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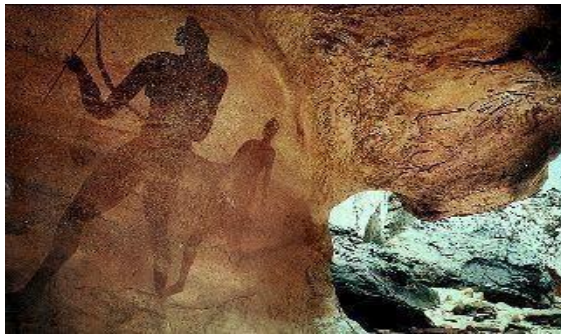
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## 8.3.1 Techniques of the Prehistoric Art

Prehistoric art, as the product of human creativity and imagination, entails painting, petrographs, pottery, and moldings. Painting and petrography are found on walls of caves and rock shelters and occasionally on more exposed surfaces of rock in almost all regions where suitable surfaces occur. One reason for this proliferation of art in Africa deserts and steppes is that in prehistoric times Africa was the most densely populated continent of the planet especially in those areas that are arid. Moreover, the dry air in those regions helped to preserve human artifacts over periods lasting for thousands of years.

### Painting:

Humans in Africa manufactured durable paints that were used to paint rocks, cave walls and pottery (**Fig. 3**). The evidence comes in the form of crucibles and pestles that have been found in settlement sites. The manufacture of these durable paints which appear



**Fig. 3. A hunter in Tassili n' Agger, Algeria, 6000 B.C. Source: British Museum, London.**

### Petrographs:

This category also includes engravings producing figurines and other forms of art as a result of etching in and out small pieces of rock until a predetermined feature is produced. This form of art was very advanced and in some cases three-dimensional figures were produced. The earliest forms of African petroglyphs are more than 6,000 years old. The production of petrographs was therefore a graphical technique.

### Pottery and Molding:

This form of art entailed the production of pots and other vessels. The main technique included coil- technique which includes interposing rows of clay on top of one another until the final vessel was produced. Additional decorations included burnishing, incision, comb- rolling on the vessel when it was still plastic dry (**fig. 4**). In some cases, human and animal figurines were molded out of clay.

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**Fig. 4. Nubian vessel (3800-3100 B.C.) decorated with a rowing boat with multiple oars, ostriches and undulating lines symbolizing water.**  
Source: Museum of Fine Arts, Boston

Masks and figures also occupy a predominant position in the prehistoric African art scenario. These were used in ceremonies conducted for religious purposes. The value of these art forms cannot be separated from the purpose for which they were used. However, no special efforts were taken to preserve the art forms. Decorative art works were used for ornamentation in textiles. Wooden art pieces were decorated with beads, shells, clay, metal, feathers and ivory.

## 8.4 WHY DID PEOPLE MAKE ART?

There are various reasons why people would want to undertake a work of art. These reasons can be assumed to be the same reasons as artists paint pictures today: to make their homes look beautiful, or in order to express some idea or feeling; or ‘playing around’ for amusement. To begin with, it may have been as a way of expressing themselves or as according to Cole, to pass idle time. He further points out that it could be speculated as to why people undertook these activities. Apart from it being an idea of passing time, it is possible to point out that they had much deeper meaning.

Some of the deeper meanings according to Cole were of two major motives. Some of the motives have been described as —wishful thinking and —sympathetic imaging. In the first instance, people will paint, draw or sketch what they would like to have, which could involve a successful hunting episode e.t.c. On the second case, the person who undertook the activity is believe that by putting down the expression would aid in achieving one’s dreams, ideas e.tc. For instance, according to Cole (1958), when an artist painted a successful hunt, it was probably because he/ she wished to have a successful hunt and by drawing it, his wish might come true.

Other artwork may have represented or commemorated an event, or an occurrence. It may have been put in place to commemorate maybe a major battle, hunt or event similar to what modern humans record by filming, drawing, or photographing daily events in our lives.

## 8.5 DATING ART

Dating art and especially rock art is not an easy task. Since rock art is stenciled on walls etc, dating their exact ages is hard to various environmental factors. The complications that come with dating the art are even harder when dealing with the African rock art because of various factors. In Europe,

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there have been climatic fluctuations over time that has made it possible to at least have an idea of

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relative ages of rock art. During the Pleistocene in Europe, there were climatic changes characterized by alternate cold glacials and warm interglacials.

With the above changes, animals that lived during the cold periods were not suited to warm conditions either moved away or died off. The woolly mammoths for example became extinct at the end of the Ice Age. Thus when paintings of animals such as the woolly mammoths are found on cave walls in Europe it is therefore known that they must have been painted before the end of the Ice Age. The variations in climate witnessed in Europe was not the same case in Africa. This makes using this form of dating in Africa difficult.

Climatic conditions in Africa did not favor rock art painting as much as it did in Europe. Due to the extreme cold climates in Europe during the late Pleistocene, humans spent most of their time in caves giving them time to paint, unlike in Africa where- by the climate was generally much warmer thus people spent less time in caves less frequently than in Africa.

## **Threats to African Art**

Many of Africa's art form representation are under threat and need to be protected. According to cultural and heritage conservation experts, these invaluable materials are particularly at risk as human populations grow and expose the materials and sites to vandalism.

## **8.6 SUMMARY**



The account of the origins of art is a very long one marked less by change than consistency. Art, as the product of human creativity and imagination, includes poetry, music, dance, and the material arts such as painting, sculpture, drawing, pottery, and bodily adornment. African art encompasses a diverse range of ceramic and cement sculptures, sculpted and carved animal bones, urban murals and rock images. The art forms depict African perspectives of the world and their ideologies.

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## 8.7 SELF- TEST QUESTIONS



1. What were the major motivating factors to pre- historic art makers?
2. Why is continental Africa possesses less cave art as compared to Europe
3. African art is older than forms of art in other continents discuss
4. Why is pre- historic art in Africa under threat?

## REFERENCES



Grine, F.; Henshilwood, S.; Sealy, C. (Jun 2000). "Human remains from Blombos Cave, South Africa: (1997-1998 excavations)". *Journal of Human Evolution* **38** (6): 755–765.

Henshilwood, C.S. 2006. Modern humans and symbolic behaviour: Evidence from Blombos Cave, South Africa. In *Origins* (ed. G. Blundell). Cape Town: Double Storey: 78–83.

Cole Sonia. 1958. *Early Man in East Africa*, 3<sup>rd</sup> ed. London: Macmillian Education Limited.

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